

Agniveer Vayu Science (Group X) - 16 Nov 2024 - Memory Based Paper

Physics

Question 1

Kirchoff's current law is based on conservation of _____ while
Kirchoff's voltage law is based on conservation of _____

Options:

- A. Mass, Charge
- B. Energy, Charge
- C. Charge, Energy
- D. Charge, Mass

Answer: C

Solution:

The correct answer is Charge, Energy.



Key Points

- A German physicist named Gustav Kirchhoff created a pair of laws in 1845 that deal with energy and current conservation in electrical circuits.
- These two laws are commonly known as Kirchhoff's Voltage and Current Law.
- Kirchhoff's current law is based on the law of conservation of charge while Kirchhoff's voltage law is based on the law of conservation of energy.
- Kirchhoff's **Current Law** is also called Kirchhoff's First Law.
 - **Kirchhoff's Current Law** states that "The algebraic sum of all currents entering and exiting a node must equal zero."
 - This law describes how a **charge** enters and leaves a wire junction point or node on a wire.
- Kirchhoff's **Voltage Law** is also called Kirchhoff's Second Law.
 - Kirchhoff's Voltage Law states that the voltage around a loop equals the sum of every voltage drop in the same loop for any closed network and equals zero.
- Kirchhoff's laws help calculate the electrical resistance of a complex network or impedance in the case of AC and the current flow in different network streams.

Question 2

The ratio of the radius of the Earth to that of the Moon is 10. The ratio of acceleration due to gravity on the Earth and on the Moon is 6. The ratio of the escape velocity from the Earth's surface to that from the Moon is

Options:

- A. 7
- B. 1.66
- C. 8
- D. 10

Answer: C

Solution:

Concept:

Acceleration due to gravity

$$g = \frac{GM}{R^2}$$

where G = Universal Gravitational constant, M = Mass of the earth, and R = Radius of the earth

Also, Escape velocity can be derived from the equation

$$v_{escape} = \sqrt{\frac{2 \times (G \times M)}{R}}$$

Putting value of G, M and R as

$$G = 6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$$

$$M = 5.972 \times 10^{24} \text{ kg}$$

$$R = 6370000 \text{ m}$$

$$v_{escape} = \sqrt{\frac{2 \times (6.67 \times 10^{-11} \times 5.972 \times 10^{24})}{6370000}}$$

$$v_{escape} = 11.18 \text{ km/s}$$

Calculations:

The ratio of acceleration due to gravity on earth to the acceleration due to gravity on the moon

$$\frac{g_{earth}}{g_{moon}} = \frac{\left(\frac{GM_E}{R_E^2}\right)}{\left(\frac{GM_M}{R_M^2}\right)}$$

$$\frac{g_{earth}}{g_{moon}} = \frac{\left(\frac{M_E}{R_E^2}\right)}{\left(\frac{M_M}{(R_E^2)/100}\right)}$$

$$6 \times 100 = \frac{M_E}{M_M}$$

$$\frac{M_E}{M_M} = 600$$

Escape velocity from the Moon of mass 1/600th of the mass of the earth and radius 1/10th radius of the earth is

$$v_{escapeMoon} = \sqrt{\frac{2 \times G \times \left(\frac{M}{600}\right)}{\left(\frac{R}{10}\right)}}$$

$$v_{escapeMoon} = \sqrt{\left(\frac{10}{600}\right) \times \frac{2 \times G \times M}{R}}$$

$$v_{escapeMoon} = \left(\frac{1}{7.75}\right) \times \sqrt{\frac{2 \times G \times M}{R}}$$

$$v_{escapeMoon} = \left(\frac{1}{7.75}\right) \times v_{escapeEarth}$$

$$\frac{v_{escapeEarth}}{v_{escapeMoon}} = 7.75 \text{ km/s}$$

Question 3

The displacement of a car during its journey is given by $s = xt^2 - yt + z$ where x, y and z are constants and t is the time. The acceleration of the car at $t = 4 \text{ s}$ is

Options:

A. 4 m/s^2

B. $2x$

C. $\sqrt{2} \text{ kmph}$

D. None of the above

Answer: B

Solution:

The correct answer is option 2) i.e. $2x$ units

CONCEPT:

- **Velocity:** The rate of change in displacement is called velocity.
- **Acceleration:** The rate of change in velocity is called acceleration.

Let the displacement of a particle be x .

Velocity, v	$\frac{dx}{dt}$
Acceleration, a	$\frac{dv}{dt}$

CALCULATION:

Given that: Displacement, $s = xt^2 - yt + z$

- The velocity of the car,

$$\Rightarrow v = \frac{ds}{dt} = \frac{d(xt^2 - yt + z)}{dt} = 2xt - y$$

- Acceleration of the car,

$$\Rightarrow a = \frac{dv}{dt} = \frac{d(2xt - y)}{dt} = 2x$$

- The acceleration of the car is time-independent and therefore, at any instant, the acceleration will be $2x$.

Question 4

In a cyclic process, the change in internal energy of the system is

Options:

- A. Positive
- B. Negative
- C. Zero
- D. Cannot be determined

Answer: C

Solution:

The correct answer is option 3) i.e. Zero

CONCEPT:

- **Cyclic process:** It is a thermodynamic process where a system comes back to its initial state i.e. the **system starts and returns to the same thermodynamic state**.
- **Internal Energy:** The internal energy of a system is identified with the random, disordered motion of molecules;
 - The total (internal) energy in a system includes potential and kinetic energy.

The internal energy of an ideal gas is calculated by

$$U = nfRT/2$$

Change in internal energy is given by:

$$\Delta U = nfR\Delta T/2$$

where n is the number of moles, f is the degree of freedom of gas, R is gas constant, and T is the temperature.

- Since the system returns to the same state after each cycle, the **change in internal energy is zero**.
- In this case, the work done will be equal to the heat energy.

EXPLANATION:

- In a cyclic process ($\Delta T = 0$ so $\Delta U = 0$), the system remains the same after each cycle. Hence, the **change in internal energy is zero**.

Question 5

A Carnot heat engine takes heat from a reservoir at 127° C and rejects heat to a sink at 27°C. Its efficiency will be

Options:

A. 20%

B. 25%

C. 30%

D. 50%

Answer: B

Solution:

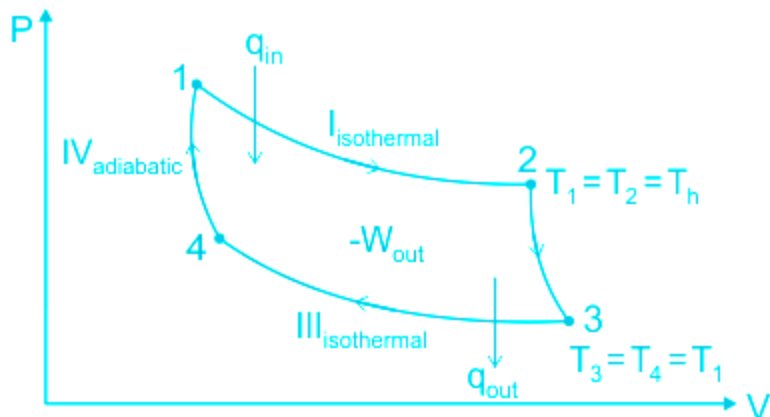
CONCEPT:

Carnot engine:

It is a theoretical thermodynamic cycle proposed by Leonard Carnot.

It gives an idea of the maximum possible efficiency that a heat engine during the conversion process of heat into work and conversely, working between two reservoirs, can possess.

Pressure-Volume diagram:



The efficiency of the Carnot cycle:

Efficiency = work done by the engine (W)/heat supplied (Q)

$$\text{Efficiency}(\eta) = 1 - T_2/T_1$$

Where, T_1 = input temperature (absolute) or source temperature, and T_2 = output temperature (absolute) or sink temperature

CALCULATION:

Given that,

$$T_1 = 127^\circ\text{C} = 273 + 127 = 400 \text{ K}$$

$$T_2 = 27^\circ\text{C} = 273 + 27 = 300 \text{ K}$$

$$\text{Efficiency}(\eta) = 1 - T_2/T_1 = 1 - (300/400) = 0.25$$

$$\text{Efficiency} = \text{work done by engine (W)/heat supplied (Q)} = 0.25 = 25 \%$$

The correct option is 2.



Additional Information

Sr. No	Isothermal Process	Adiabatic Process

1.	It is a thermodynamic process in which the temperature of a system remains constant	It occurs without transferring heat or mass between a thermodynamic system and its surrounding
2.	The transfer of heat into or out of the system happens so slowly that thermal equilibrium is maintained .	Unlike an isothermal process, an adiabatic process transfers energy to the surroundings only as work

Question 6

Dimensional formula of ω in equation $y = a \sin (\omega t + kx)$ is

Options:

A. $M^0 L^0 T^{-1}$

B. $M^0 L^{-1} T^0$

C. $M L^0 T^0$

D. $M^0 L T^{-1}$

Answer: A

Solution:

CONCEPT:

Dimensions:

- **Dimensions** of a physical quantity are the powers to which the fundamental units are raised to obtain one unit of that quantity.

EXPLANATION:

Given $y = a \sin (\omega t + kx)$

- Since the **trigonometric function is a dimensionless quantity**. So, $(\omega t + kx)$ is also a **dimensionless quantity**.

Here, t = time

- The dimension of t is given as,

$$\Rightarrow [t] = [M^0 L^0 T^1] \text{ -----(1)}$$

- Since the quantity ωt is dimensionless.

$$\Rightarrow [\omega t] = [M^0 L^0 T^0] \text{ -----(2)}$$

Let the dimension of ω as,

$$\Rightarrow [w] = [M^x L^y T^z] \text{ -----(3)}$$

by equation 1, equation 2 and equation 3,

$$\Rightarrow [M^0 L^0 T^1] [M^x L^y T^z] = [M^0 L^0 T^0]$$

$$\Rightarrow [M^x L^y T^{z+1}] = [M^0 L^0 T^0]$$

$$\Rightarrow x = 0$$

$$\Rightarrow y = 0$$

$$\Rightarrow z + 1 = 0$$

$$\Rightarrow z = -1$$

- So dimensional formula for w is,

$$\Rightarrow [w] = [M^0 L^0 T^{-1}]$$

- Hence, option 1 is correct.

Question 7

Magnetic field at the centre of a circular coil of radius R due to current I flowing through it is B . The magnetic field at a point along the axis at distance R from the centre is _____

Options:

A. B

B. $B/\sqrt{2}$

C. $B/\sqrt{4}$

D. $B/\sqrt{8}$

Answer: D

Solution:

CONCEPT:

The magnetic field at point O due to the circular coil is given by:

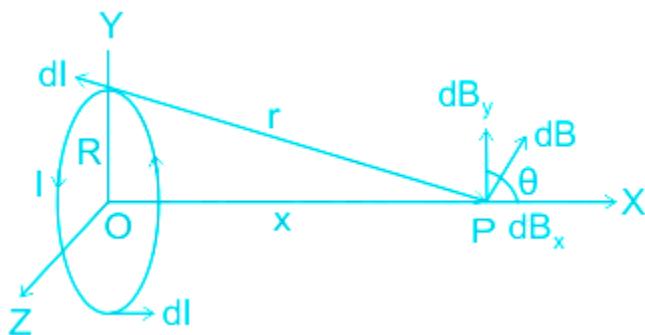
$$B = \frac{\mu_0 I}{2R}$$

where B is the magnetic field at the center, μ_0 permeability of the medium I is the current in the circular loop, and R is the radius of the circular coil.

- The magnetic field on the axis of a circular loop at a distance of x from the center of wire of radius R:

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

where I is the current in the loop, x is the distance from the center, R is the radius of the circular loop.



CALCULATION:

Given that $x = R$; and Magnetic field at the center of a circular coil of radius R due to current I flowing through it is B.

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

$$B' = \frac{\mu_0 I R^2}{2(R^2 + R^2)^{3/2}} = \frac{\mu_0 I R^2}{2(2R^2)^{3/2}} = \frac{\mu_0 I R^2}{2\sqrt{8}(R)^3} = \frac{\mu_0 I}{\sqrt{8}(2R)}$$

We know that The magnetic field at point O due to the circular coil is given by:

$$B = \frac{\mu_0 I}{2R}$$

$$\text{So } B' = \frac{\mu_0 I}{\sqrt{8}(2R)} = \frac{B}{\sqrt{8}}$$

Hence the correct answer is **option 4**.

Question 8

Determine the magnitude of induced EMF (in V) in a coil, if the current changes from +2 to -2A in 0.5 second and the coefficient of mutual induction is 0.5

Options:

A. 2

B. 4

C. 3

D. 6

Answer: B

Solution:

Concept:

The coefficient of mutual induction of two coils is numerically equal to the magnetic flux linked with one coil when unit current flows through the neighbouring coil.

If E is the induced emf in the coil at any instant of time, then from the laws of electromagnetic induction:

$$E = -\frac{d\phi}{dt} \text{ ----(1)}$$

$$\phi = M \times I$$

M is the coefficient of mutual induction

I is the current.

Equation (1) becomes:

$$E = -\frac{d(MI)}{dt} = -M \frac{di}{dt}$$

Calculation:

Putting the respective values, we get:

$$E = -\frac{0.5(-2-2)}{0.5} = 4 \text{ V}$$

Question 9

An AC current is expressed as $i = 50 \sin 100 t$ A. What is the half-cycle average value of that current?

Options:

A. $\frac{50}{\pi}$ A

B. 50 A

C. 100 A

D. $\frac{100}{\pi}$ A

Answer: D

Solution:

Concept:

- The **root mean square current /voltage (rms)** of an AC circuit is the **effective current/voltage** of that circuit.
- The maximum value of the potential in an AC circuit is called the **peak value of voltage**.
- The average value of the potential and current of an AC circuit for a cycle is called **average potential and average current**.

Average current over half cycle $(I_{avg}) = \frac{2 I}{\pi}$

Average current over complete cycle = 0

rms current $(I_{rms}) = \frac{I}{\sqrt{2}}$

Where I is maximum/peak current in the circuit.

Calculation:

Given that: $i = 50 \sin 100 t$ A

$\therefore$ The peak value of current, $I = 50$ A

Average current over half cycle = Average value of sinusoidal current:

$\Rightarrow \frac{2 I}{\pi} = \frac{2 \times 50}{\pi}$ A

$\therefore$ **Average current over half cycle** $= \frac{100}{\pi}$ A

Average current over complete cycle = 0 A



Mistake Points

In the given question AC (Alternating Current) wave is mentioned, which consists of peak values ' I ' in the positive half cycle and ' $-I$ ' in the negative half cycle.

$\therefore$ **Average current over the complete cycle (positive + negative half cycle) = 0 ampere**

That's why the **average value of AC wave is calculated over half cycle**, which is equivalent to the full-wave rectifier's average value $= \frac{2I}{\pi}$

The average value for a half and full-wave rectifier are:

$$I_{avg} = \frac{I}{\pi} \text{ for half-wave rectifier}$$

$$I_{avg} = \frac{2I}{\pi} \text{ for full-wave rectifier}$$

Note that, half wave rectifier consists of the one-half cycle and inactive for other half cycle. So that its average value is half of the full-wave rectifier's average value.

Where the average value of AC wave is calculated over half cycle is equal to full-wave rectifier's average value.

Question 10

The turns ratio of a transformer used in half-wave rectifier is $n_1 : n_2 = 12 : 1$. The primary is connected to the power mains of 220 V, 50 Hz. Assuming the diode resistance in forward bias to be zero, calculate the dc voltage across the load

Options:

A. 8.24 V

B. 4.12 V

C. 2.8 V

D. 16.48 V

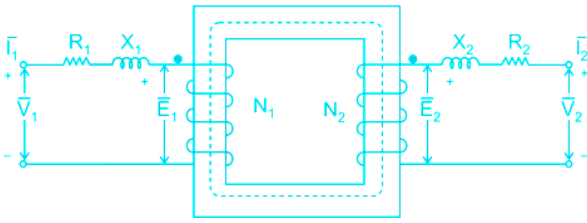
Answer: A

Solution:

Concept:

Transformer:

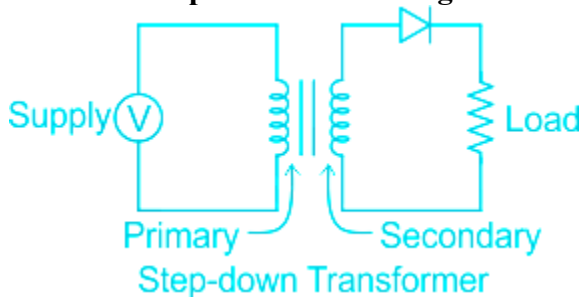
- It is basically a voltage control device that is used widely in the distribution and transmission of alternating current power.
- A device used in the power transmission of electric energy.



- The transformer works on the principle of **Faraday's law of electromagnetic induction and mutual induction**.
- In the transformer, $\frac{N_1}{N_2} = \frac{I_2}{I_1}$, where N_1 and N_2 = the number of turns in the primary and secondary coil respectively, I_1 and I_2 = the current in the primary and secondary coil respectively.
- In terms of voltage, $\frac{N_1}{N_2} = \frac{V_1}{V_2}$ where, N_1 and N_2 = the number of turns in the primary and secondary coil respectively, V_1 and V_2 = the voltage in the primary and secondary coil respectively.

Half-wave rectifier:

- Half-wave rectifiers transform AC voltage to DC voltage.
- A halfwave rectifier circuit uses only one diode for the transformation.
- **A halfwave rectifier is defined as a type of rectifier that allows only one-half cycle of an AC voltage waveform to pass while blocking the other half cycle.**



- The DC output voltage, $V_{DC} = \frac{V_m}{\pi} = \frac{\sqrt{2}V_2}{\pi}$

Calculation:

Given,

The turns ratio of a transformer used in a half-wave rectifier is $N_1 : N_2 = 12 : 1$

The voltage at the primary coil, $V_1 = 220 \text{ V}$

The frequency, $f = 50 \text{ Hz}$

We know that, $\frac{N_1}{N_2} = \frac{V_1}{V_2}$

$$V_2 = V_1 \frac{N_2}{N_1}$$

$$V_2 = 220 \times \frac{1}{12} = 18.33 \text{ V}$$

The DC output voltage is calculated as, $V_{DC} = \frac{V_m}{\pi} = \frac{\sqrt{2}V_2}{\pi}$

$$V_{DC} = \frac{\sqrt{2} \times 18.33}{3.14} = 8.255V$$

Additional Information

- **Faraday's law of electromagnetic induction:**
 - Faraday's Laws of Electromagnetic Induction consist of two laws.
 - **Faraday's first law of electromagnetic induction:** States **Whenever a conductor is placed in a varying magnetic field, an electromotive force is induced**
 - **Faraday's second law of electromagnetic induction:** States that the induced emf is equal to the rate of change in magnetic flux with respect to time.
 - Formula-induced emf, $e = -N \frac{\Delta\phi}{\Delta t}$ where N = number of turns, $\Delta\phi = BA\cos\theta$ = Magnetic flux
- **Mutual induction:**
 - It is defined as **the property of the coils that enables it to oppose the changes in the current in another coil.**
 - With a change in the current of one coil, the flow changes too thus inducing EMF in the other coil

Question 11

The excess pressure inside a soap bubble is thrice the excess pressure inside a second soap bubble. The ratio between the volume of the first and the second bubble is :

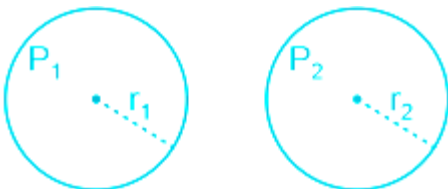
Options:

- A. 1 : 9
- B. 1 : 3
- C. 1 : 81
- D. 1 : 27

Answer: D

Solution:

Calculation:



$$\Rightarrow P_1 - P_0 = \frac{4T}{r_1}$$

$$\Rightarrow P_2 - P_0 = \frac{4T}{r_2}$$

$$\Rightarrow P_1 - P_0 = 3(P_2 - P_0)$$

$$\Rightarrow \frac{4T}{r_1} = 3 \frac{4T}{r_2}$$

$$\Rightarrow r_2 = 3r_1$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{1}{27}$$

$\therefore$ the ratio between the volume of the first and second bubble is 1:27.

Question 12

A current-carrying straight conductor is placed in a magnetic field. The conductor experiences the maximum force when the angle between the direction of the current in it and the direction of the magnetic field is:

Options:

A. 45°

B. 30°

C. 90°

D. 60°

Answer: C

Solution:

The correct answer is 90° .



Key Points

- The **force** experienced by a **current-carrying conductor** placed in a **magnetic field** is given by the formula **$F = BIL\sin\theta$** , where B is the magnetic field strength, I is the current flowing through the conductor, L is the length of the conductor, and θ is the **angle** between the direction of the current and the direction of the magnetic field.
- The **maximum force** is experienced when the angle is **90°** , but in this case, the **$\sin\theta$** becomes **1**, which is the maximum value.

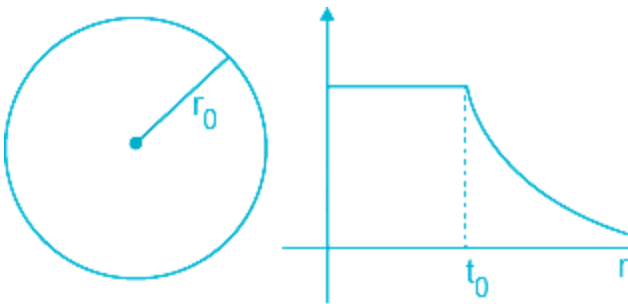
- However, in practical scenarios, it is **not always** possible to have the conductor **perpendicular** to the magnetic field.
- When the angle is **0 degrees**, the force experienced is **zero** because $\sin 0$ is zero.
- When the angle is **180 degrees**, the force experienced is **also zero** because $\sin 180$ is again zero.
- Therefore, the maximum force is experienced when the angle between the direction of the current in the conductor and the direction of the magnetic field is **90 degrees** because the $\sin 90$ is the maximum value of 1.

Additional Information

- When the angle between the direction of the current in the conductor and the direction of the magnetic field is **30 degrees**, the $\sin \theta$ is 0.5, which is **less than** the maximum value of 1.
 - Therefore, the force experienced will be **less than** the maximum force.
- When the angle between the direction of the current in the conductor and the direction of the magnetic field is **60 degrees**, the $\sin \theta$ is 0.866, which is less than the maximum value of 1.
 - Therefore, the force experienced will be **less** than the maximum force.

Question 13

The given graph shows variation (with distance r from centre) of



Options:

- A. Electric field of a uniformly charged spherical shell
- B. Potential of a uniformly charged spherical shell
- C. Electric field of a uniformly charged sphere
- D. Potential of a uniformly charged sphere

Answer: B

Solution:

Concept:

For a uniformly charged spherical shell, electric potential inside it is given by

$$V_{inside} = V_{surface} = \frac{kq}{r_0} = \text{constant}$$

(where r_0 = radius of the shell)

Electric potential outside the shell at a distance r is

$$V_{outside} = \frac{kq}{r}$$

$$V \propto \frac{1}{r}$$

$\therefore$ The given graph represents the variation of r and potential of a uniformly charged spherical shell. The potential increases with decrease in r and r increases with decrease in the potential V .

Question 14

A concave lens with unequal radii of curvature made of glass ($\mu_g = 1.5$) has a focal length of 40 cm. If it is immersed in a liquid of refractive index $\mu_l = 2$, then

Options:

- A. it behaves like convex lens of 80 cm focal length
- B. it behave like a convex lens of 20 cm focal length
- C. its focal length becomes 60 cm
- D. nothing can be said

Answer: A

Solution:

Concept:

The lens maker's formula relates the focal length of a lens to its radii of curvature and the refractive indices of the lens material and the medium in which the lens is placed.

Lens maker's Formula

For a lens in air, the lens maker's formula is:

$$\frac{1}{f} = (\mu_{\text{lens}} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

where:

f is the focal length of the lens,

μ_{lens} is the refractive index of the lens material,

R_1 and R_2 are the radii of curvature of the two lens surfaces.

When the lens is immersed in a liquid with refractive index μ_{liquid} , the effective focal length f' is given by:

$$\frac{1}{f'} = \left(\frac{\mu_{\text{lens}}}{\mu_{\text{liquid}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Explanation:

Given Data

Refractive index of the glass (lens material), $\mu_{\text{glass}} = 1.5$,

Focal length of the lens in air, $f = -40$ cm (negative for concave lens),

Refractive index of the liquid, $\mu_{\text{liquid}} = 2$.

Calculations

The Lens maker's Formula for Air :

$$\frac{1}{f} = (\mu_{\text{glass}} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \frac{1}{-40} = (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \frac{1}{-40} = 0.5 \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{-40 \times 0.5} = -\frac{1}{20}$$

The Effective Focal Length in the Liquid :

$$\frac{1}{f'} = \left(\frac{\mu_{\text{glass}}}{\mu_{\text{liquid}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \frac{1}{f} = \left(\frac{1.5}{2} - 1 \right) \left(-\frac{1}{20} \right) \frac{1}{f} = (0.75 - 1) \left(-\frac{1}{20} \right) \frac{1}{f} = (-0.25) \left(-\frac{1}{20} \right) \frac{1}{f} = \frac{0.25}{20}$$

$$\frac{1}{f'} = \frac{1}{80}$$

Thus, the focal length of the lens when immersed in the liquid is:

$$f' = 80 \text{ cm}$$

When the concave lens is immersed in a liquid with a refractive index of 2, its focal length will become 80 cm .

The correct option is (1).

Question 15

If the empirical formula for the observed wavelengths (λ) for hydrogen is $1/\lambda = R(1/4^2 - 1/n^2)$, where 'n' is integral values higher than 4, then it represents the _____ series.

Options:

A. Balmer

B. Brackett

C. Paschen

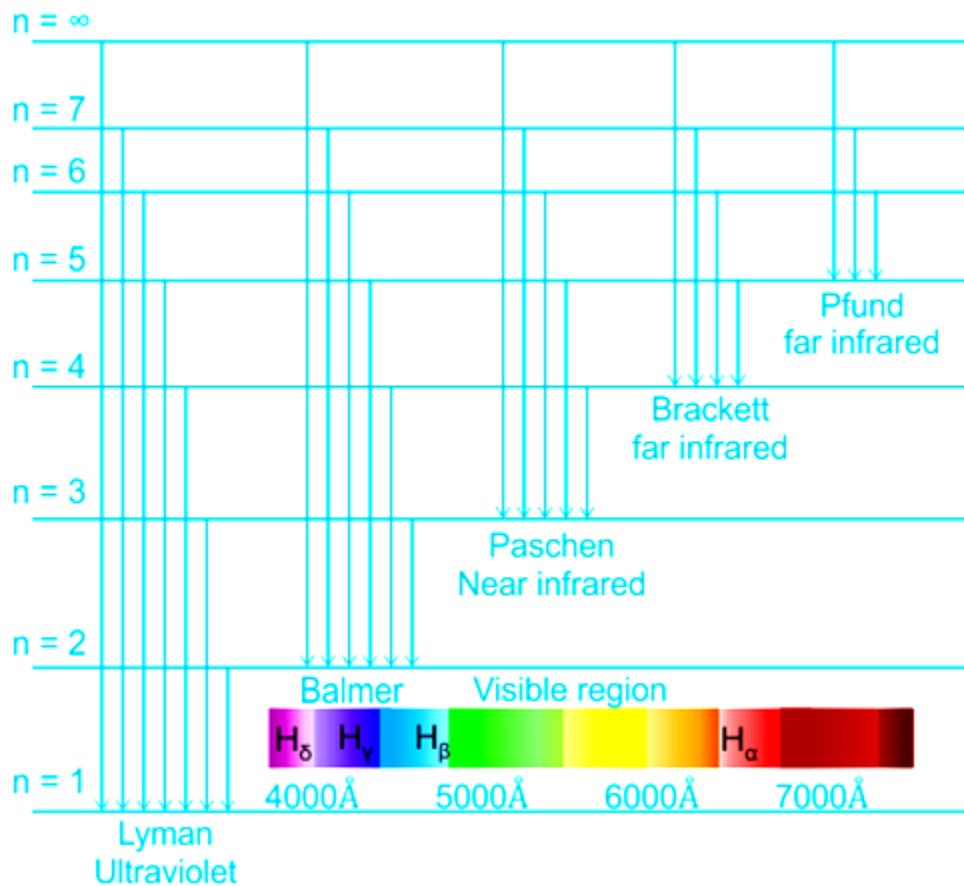
D. Lyman

Answer: B

Solution:

CONCEPT:

- **Hydrogen Spectrum:** The hydrogen spectrum is a quantized electronic structure of an atom.
 - The energetically excited hydrogen atoms released electromagnetic radiation.
 - The hydrogen emission spectrum comprises radiation of discrete frequencies.



- The atomic spectrum of hydrogen consists of several lines that have been grouped into 5 series: **Lyman, Balmer, Paschen, Brackett, and Pfund.**

The wavelengths (λ) for hydrogen is given by:

$$1/\lambda = R(1/m^2 - 1/n^2)$$

Where m and n are the principle quantum numbers, R is rydberg constant.

EXPLANATION:

For **Brackett series:**

Principle quantum number (n) = 4

So $1/\lambda = R(1/4^2 - 1/n^2)$. Hence option 2 is correct.

Question 16

Zener diode is a _____ doped PN junction diode and connected in _____ bias in the circuit.

Options:

- A. lightly, forward
- B. heavily, forward
- C. lightly, reversed
- D. heavily, reversed

Answer: D

Solution:

Concept:

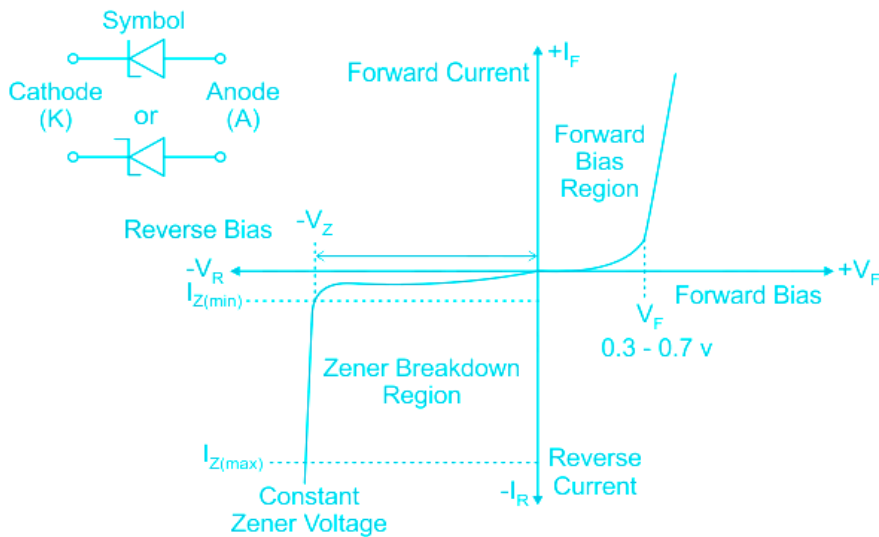
Zener Diode:

- A Zener diode not only allows current to flow from the anode to the cathode but also, in the reverse direction on reaching the Zener voltage.
- Due to this functionality, Zener diodes are the most commonly used semiconductor diodes.
- It is a heavily doped semiconductor device that is designed to operate in the reverse direction.

There are two types of breakdowns for a Zener Diode

- **Avalanche Breakdown in Zener diode:**
 - Avalanche breakdown occurs in the normal diode and Zener Diode at high reverse voltage.
 - When a high value of reverse voltage is applied to the PN junction, the free electrons gain sufficient energy and accelerate at high velocities.
- **Zener Breakdown in Zener diode:**
 - When the applied reverse bias voltage reaches closer to the Zener voltage, the electric field in the depletion region gets strong enough to pull electrons from their valence band.
 - The valence electrons that gain sufficient energy from the strong electric field of the depletion region break free from the parent atom. At the Zener breakdown region, a small increase in the voltage results in the rapid increase of the electric current.

Symbol and VI characteristic:



Explanation:

The Zener diode is a heavily doped PN junction and is connected in reverse biased in the circuit.

Question 17

The refractive index of glass is 1.5 for light whose wavelength in a vacuum is 6000 Å. The wavelength of this light when it passes through glass is

Options:

- A. 4000 Å
- B. 6000 Å
- C. 9000 Å
- D. 15000 Å

Answer: A

Solution:

Concept:

Refractive index:

- It is the ratio between the speed of light in air to the speed in a medium.
- Formula, refractive index, $\mu = \frac{c}{v}$

- In terms of the wavelength, it is equal to the ratio of the wavelength of light in air to the wavelength of light in the medium.
- Formula, $\mu = \frac{\lambda_0}{\lambda}$ where, λ_0 = wavelength in the air, λ = wavelength in the medium.

Calculation:

Given,

The refractive index, $\mu = 1.5$

The wavelength in the vacuum, $\lambda_0 = 6000 \text{ \AA}$

We know that, $\mu = \frac{\lambda_0}{\lambda}$

$$\lambda = \frac{\lambda_0}{\mu}$$

$$\lambda = \frac{6000}{1.5}$$

$$\lambda = 4000 \text{ \AA}$$

Question 18

A source of sound of frequency 600 Hz is placed inside water. The speed of sound in water is 1500 m/s and in air it is 300 m/s. The frequency of sound recorded by an observer in air is:

Options:

A. 3000 Hz

B. 1500 Hz

C. 600 Hz

D. 120 Hz

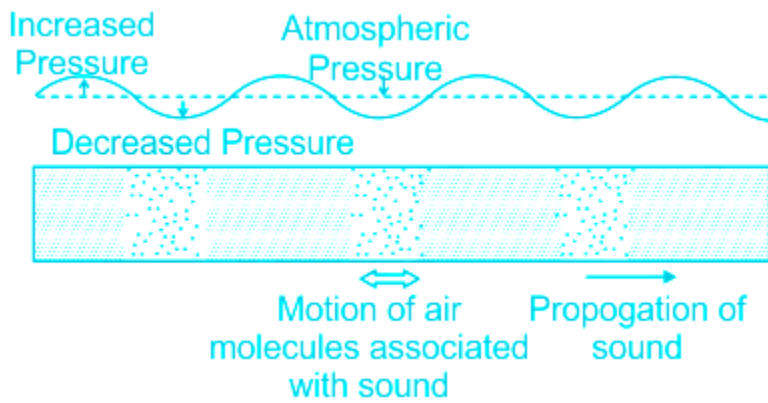
Answer: C

Solution:

Concept:

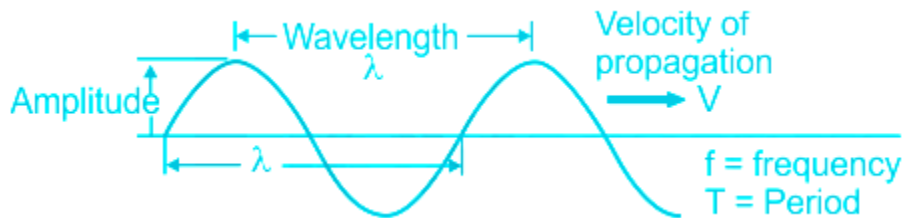
- **Sound is generated** due to **oscillations of object** Ex: vocal cord in our throat
- In this **vibrating molecules transfer energy** to the neighbouring molecule for transmission.
- It is also known as a **pressure wave** as it forms **alternate high and low-pressure regions**.

- It is **longitudinal wave** as particles oscillate **parallel to the direction of propagation**.



Explanation:

- Sound travels as a wave in medium therefore its properties depend on wavelength, frequency, and velocity.



- **Wavelength(λ):** The **distance** traveled by wave **within one complete oscillation**.
- **Frequency(ν):** The **number of vibrations** completed **in one second**.
- Frequency can be also given by **reciprocal of a time period(T)**.
- **Velocity(v):** The distance traveled by the wave **per unit time**.

$$Velocity = \frac{Distance}{Time} = \frac{Wavelength(\lambda)}{Time(T)}$$

As we know, $\nu = \frac{1}{T}$ therefore equation of velocity can be given as,

$$V = \lambda \times \nu$$

- When a wave travels in the **same medium** its **velocity** remains **constant**.
- When a **wave travels from one medium to another**, then **velocity and wavelength change**.
- But the **frequency** of a wave remains **constant**.
- As the **frequency** is a **characteristic of a source** i.e how much it is vibrating per unit time.
- Therefore frequency will be the same option 3 is correct.



Mistake Points

- **Doppler effect** is an **apparent change in frequency** due to the **motion of the observer or source**.
- **Relative approach:** one of the source and listener, or both **moving towards another**.
- The **apparent frequency is greater than the real frequency** of the source.
- **Relative recede:** one of the source and listener, or both **moving away from another**.
- The **apparent frequency is less than the real frequency** of the source.



Additional Information

- Speed of sound in different mediums.

Medium	Speed of sound (m/s)
Air	343
Water	1480
Mercury	1452
Glass	5640
Aluminum	6420
Diamond	12000
Helium	965
Oxygen	316

Question 19

The density of a material in SI units is 128 kg m^{-3} . In certain units in which the unit of length is 25 cm and the unit of mass is 50 g, the numerical value of density of the material is

Options:

- A. 40
- B. 16
- C. 640
- D. 410

Answer: A

Solution:

Concept:

Given,

Density of the material = 128 kg m^{-3}

Mass of the material = 50 g

Length of the material = 25 cm

To convert a measured value from one system to another system, we use

$$\Rightarrow N_1 u_1 = N_2 u_2$$

Where N is the numeric value and 'u' is the unit.

Calculation:

We get,

$$\Rightarrow 128 \frac{kg}{m^3} = N_2 \frac{50g}{(25\text{ cm})^3}$$

$$[density = \frac{mass}{volume}]$$

$$\Rightarrow \frac{128 \times 1000g}{100 \times 100 \times 100 \text{ cm}^3} = \frac{N_2 \times 50g}{25 \times 25 \times 25 \text{ cm}^3}$$

$$\Rightarrow N_2 = \frac{128 \times 1000}{100 \times 100 \times 100} \times \frac{25 \times 25 \times 25}{50}$$

$$\Rightarrow N_2 = \frac{128}{1000} \times \frac{25 \times 25}{2}$$

$$\Rightarrow N_2 = \frac{128}{8} \times \frac{5}{2}$$

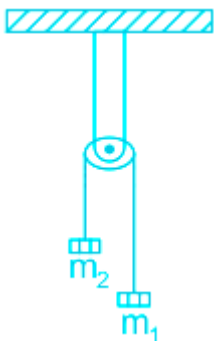
$$\Rightarrow N_2 = \frac{64}{8} \times 5 = 8 \times 5$$

$$\therefore N_2 = 40$$

Therefore, the numerical value of the density of the material is 40.

Question 20

Two masses m_1 and m_2 are attached to a string which Passes over a frictionless fixed pulley as is shown in the figure, If $m_1 = 7\text{Kg}$, $m_2 = 3\text{kg}$, and $g = 10 \text{ ms}^{-2}$, the acceleration of the masses is:



Options:

A. 2.1 ms^{-2}

B. 4ms^{-2}

C. 42ms^{-2}

D. 10ms^{-2}

Answer: B

Solution:

CONCEPT:

- **Tension in String:** The reaction force acting on a **tight string** when it is holding something is called tension in the string.
- **Pulley:** Pulley is the type of simple machine used to lift any weight by applying lesser force.
 - When not mentioned, the mass of pulley and string is **considered to be negligible**.
- **Freebody Diagram:** The free body diagram of an object **represents all forces acting with their directions**.
 - When two objects are connected to the pulley as shown in the figure then we assume the body with **more mass is descending with acceleration** and the body with lesser mass is getting uplifted with the **same acceleration**.
- Acceleration and Tension offered by the **massless Pulley and string** are the **same throughout the string**.
- The **weight of an object** is the product of its **mass and acceleration** due to gravity.

EXPLANATION:

- FBD of all two block



- As the pulley is frictionless so the magnitude of the acceleration of both masses is equal.

From FBD,

$$\text{For } m_1: m_1 a = m_1 g - T \dots (1)$$

$$\text{For } m_2: m_2 a = T - m_2 g \dots (2)$$

Adding equation (1) + (2)

$$a (m_1 + m_2) = g (m_1 - m_2)$$

$$\text{or } a = \frac{g(m_1 - m_2)}{(m_1 + m_2)} = \frac{10 \times (7 - 3)}{7 + 3} = 4$$

The acceleration of the masses is 4 m/s^2 .

Question 21

A uniform chain of length L and mass m is lying on a smooth table and one-third of its length is hanging vertically down over the edge of the table. If g is the acceleration due to gravity, then work required to pull the hanging part on the table will be:

Options:

A. $\frac{MgL}{6}$

B. $\frac{MgL}{9}$

C. $\frac{MgL}{12}$

D. $\frac{MgL}{18}$

Answer: D

Solution:

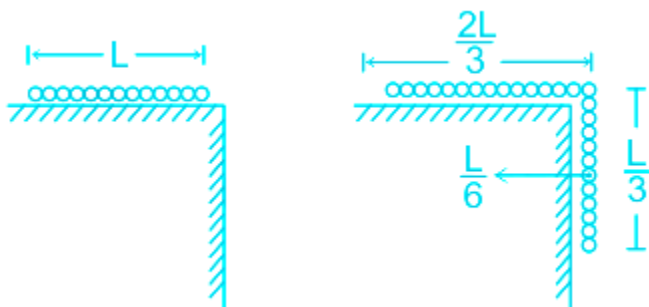
Concept:

- **Work is said to be done** by a force when the body is displaced actually through some distance in the direction of the applied force.
- Since the **body is being displaced in the direction of F** , therefore work done by the force in displacing the body through a distance s is given by:

$$W = \vec{F} \cdot \vec{s}$$

Or, $W = Fs \cos \theta$

Calculation:



Total mass m & length $L = mg$

Weight of the hanging part $(L/3) = 1/3 \text{ mg}$

This weight acts at the centre of gravity of the hanging part, which is at a distance of $= L/6$ from the table

We know that,

Work done = Force $\times$ Distance

$$Work\ done = \frac{mg}{3} \times \frac{l}{6} = \frac{MgL}{18}$$

Question 22

A ball of mass M moving with a speed of 2 m/s hits another ball of mass 1 kg moving in the same direction with a speed of 1 m/s. If the kinetic energy of center of mass is $\frac{4}{3}$ Joule, then the magnitude of M is

Options:

A. 1 kg

B. 0.25 kg

C. 0.50 kg

D. 2 kg

Answer: C

Solution:

CONCEPT:

As we know;

$$v_{CM} = \frac{m_1v_1 + m_2v_2}{m_1 + m_2}$$

CALCULATION:

Given: mass $m_1 = M$

$m_2 = 1\text{kg}$

As we have;

$$v_{CM} = \frac{m_1v_1 + m_2v_2}{m_1 + m_2}$$

$$\Rightarrow v_{CM} = \frac{2M+1}{M+1}$$

The kineticsof the center of mass is $\frac{4}{3}$ Joule, therefore we have;

$$\frac{4}{3} = \frac{1}{2}(M + 1)\left(\frac{2M+1}{M+1}\right)^2$$

Now solving the above equation we have;

$$M = 0.50 \text{ kg}$$

Hence option 3) is the correct answer.

Question 23

Consider a circular ring of mass 1 kg and diameter 0.2 m. It is making 10 rotations per second about an axis passing through its centre and normal to the surface. The value of angular momentum is:-

Options:

A. $0.628 \text{ kg m}^2/\text{sec}$

B. $0.4 \text{ kg m}^2/\text{sec}$

C. $1.256 \text{ kg m}^2/\text{sec}$

D. $0.2 \text{ kg m}^2/\text{sec}$

Answer: A

Solution:

Concept:

- Angular momentum (**L**) or rotational momentum is the **moment of inertia times (I)angular velocity (ω)**.
- Angular momentum is the Vector quantity and its directly proportional to the toque applied to the body.
- When the direction of torque and direction of rotation is the same then angular momentum increases and when the direction of torque and rotation is opposite then angular momentum decreases.

$$\therefore \mathbf{L} = \mathbf{I} \times \boldsymbol{\omega}$$

Where,

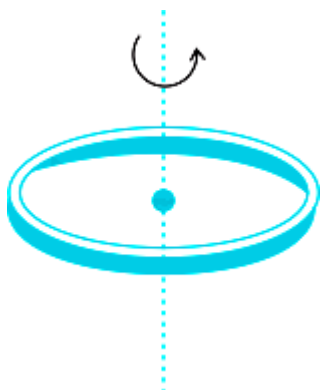
L is angular momentum
and, ω is angular velocity

Explanation:

Given data,

Circular ring of mass (**m**) = 1 kg,

diameter (d) = 0.2 m $\rightarrow$ radius (**r**) = 0.1 m



- As we know the angular momentum of a ring for which axes of rotation pass from its center and normal to the plane is $\mathbf{I} = \mathbf{mr}^2$
- As the ring rotates 10 revolutions per second it means it has angular velocity (ω) = $10 \times 2\pi \frac{\text{rad}}{\text{s}}$

Now Angular momentum (**L**) = $\mathbf{I} \times \omega$

$$\therefore \mathbf{L} = \mathbf{mr}^2 \times \omega$$

$$\therefore \mathbf{L} = (1) \cdot (0.1)^2 \times 20\pi$$

$$\therefore \mathbf{L} = 0.628 \text{ kg m}^2\text{s}^{-1}$$

Hence the angular momentum of the circular ring becomes $0.628 \text{ kg m}^2\text{s}^{-1}$.

Note: This question was deleted in RSMSSB Lab Assistant 2016 Exam

Question 24

Two plane mirrors are placed making angle of 30° with each other. Find the number of images formed if we place an object between the mirrors asymmetrically.

Options:

A. 15

B. 12

C. 10

D. 11

Answer: D

Solution:

CONCEPT:

- **Number of images:** When two mirrors are inclined at an angle θ then the number of images formed is given by the below formula.

$$n = \frac{360}{\theta} - 1$$

CALCULATION:

Given that $\theta = 30^\circ$ symmetrically

- The number of images formed if we place an object between the mirrors asymmetrically is.

$$\Rightarrow n = \frac{360}{\theta} - 1$$

$$\Rightarrow n = \frac{360}{30} - 1 = 12 - 1 = 11$$

$$\Rightarrow n = 11$$

- So the correct answer is **option 2**.
-

Question 25

The coefficient of areal expansion of a material is $1.6 \times 10^{-5} \text{ K}^{-1}$. Which one of the following gives the value of coefficient of volume expansion of this material?

Options:

A. $0.8 \times 10^{-5} \text{ K}^{-1}$

B. $2.4 \times 10^{-5} \text{ K}^{-1}$

C. $3.2 \times 10^{-5} \text{ K}^{-1}$

D. $4.8 \times 10^{-5} \text{ K}^{-1}$

Answer: B

Solution:

CONCEPT:

Thermal expansion: When the temperature of any object is changed then the tendency of changing the shape, area, and volume of the body is called the **thermal expansion** of that object.

There are three types of thermal expansion:

1. Linear thermal expansion: When the temperature of anybody is changed then the tendency of changing the length of the body is called a linear thermal expansion of that body.

Change in length is defined as;

$$\Delta L = L \alpha \Delta T$$

Where ΔL = changes in length, L = original length, ΔT = changes in temperature of the body, and α = coefficient of linear expansion.

2. Areal Expansion: When the temperature of anybody is changed then the tendency of changing the area of the body is called the areal thermal expansion of that body.

The change in area is defined as;

$$\Delta A = A \beta \Delta T$$

Where ΔA = changed in the area, A = the original area, β = the coefficient of areal expansion, and ΔT = temperature change.

3. Volume expansion: When the temperature of anybody is changed then the tendency of changing the volume of the body is called a volume thermal expansion of that body.

Change in volume (ΔV) is given by:

$$\Delta V = V \gamma \Delta T$$

Where V = original volume, γ = the coefficient of volume expansion and ΔT = temperature change.

The relation between α , β and γ is given by:

$$\beta = 2\alpha, \text{ and } \gamma = 3\alpha$$

EXPLANATION:

$$\beta = 2\alpha, \text{ and } \gamma = 3\alpha$$

$$\text{So } \gamma = 3\beta/2 = 1.5 \beta$$

Given that, **coefficient of areal expansion** (β) = $1.6 \times 10^{-5} \text{ K}^{-1}$

Coefficient of volume expansion (γ) = $1.5 \times 1.6 \times 10^{-5} \text{ K}^{-1} = 2.4 \times 10^{-5} \text{ K}^{-1}$

Question 26

Calculate $\int (x^2 \cos x) dx$

Options:

A. $x^2 \sin x + 2x \cos x - \sin x + C$

B. $x^2 \sin x - 2x \cos x + \sin x + C$

C. $x^2 \sin x + 2x \cos x + 2 \sin x + C$

D. $x^2 \sin x + 2x \cos x - 2 \sin x + C$

Answer: D

Solution:

Concept:

Integral property:

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C; n \neq -1$
- $\int \frac{1}{x} dx = \ln x + C$
- $\int e^x dx = e^x + C$
- $\int a^x dx = (a^x / \ln a) + C; a > 0, a \neq 1$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$

Integration by parts: Integration by parts is a method to find integrals of products. The formula for integrating by parts is given by:

$$\Rightarrow \int u v dx = u \int v dx - \int \left(\frac{du}{dx} \times \int v dx \right) dx + C$$

where u is the function $u(x)$ and v is the function $v(x)$

ILATE rule is Usually, the preference order of this rule is based on some functions such as Inverse, Logarithm, Algebraic, Trigonometric and Exponent.

Calculation:

$$I = \int (x^2 \cos x) dx$$

Using integration by parts

$$\Rightarrow I = x^2 \int \cos x dx - \int \left(\frac{dx^2}{dx} \times \int \cos x dx \right) dx + C$$

$$\Rightarrow I = x^2 \sin x - \int (2x \sin x) dx + C$$

$$\Rightarrow I = x^2 \sin x - 2 \int x \sin x dx + C$$

$$\Rightarrow I = x^2 \sin x - 2 \left[x \int \sin x dx - \int \left(\frac{dx}{dx} \times \int \sin x dx \right) dx \right] + C$$

$$\Rightarrow I = x^2 \sin x - 2 \left[-x \cos x - \int (-\cos x) dx \right] + C$$

$$\Rightarrow I = x^2 \sin x + 2x \cos x - 2 \sin x + C$$

Question 27

Find the value of $\det(3A)$ for the following matrix:

$$A = \begin{bmatrix} 4 & 7 & 1 \\ -1 & 3 & 2 \\ -2 & 0 & 5 \end{bmatrix}$$

Options:

A. 1458

B. 81

C. 27

D. 1971

Answer: D

Solution:

Concept:

1. Determinant of a 3×3 matrix:

- Let A be a 3×3 matrix given by:

$$A = \begin{bmatrix} a & b & c \\ f & e & d \\ g & h & i \end{bmatrix}$$

then the value of $|A|$ also written as $\det(A)$ is:

$$\det(A) = a(ei - dh) - b(fi - dg) + c(fh - eg)$$

2. Property of determinant of a matrix:

- Let A be a matrix of order $n \times n$ and $\det(A) = k$. Then for a scalar c , the following property holds:

$$\det(cA) = c^n \det(A)$$

Calculation:

First evaluate the determinant of the given matrix:

$$\det(A) = 4(15 - 0) - 7(-5 + 4) + 1(0 + 6)$$

$$= 4(15) - 7(-1) + 1(6)$$

$$= 60 + 7 + 6$$

$$= 73$$

Now using the property the value of $\det(3A)$ is:

$$\det(3A) = 3^3 \det(A)$$

$$= 27 \times 73$$

$$= 1971$$

Question 28

$${}^n P_5 = 42 \times {}^n P_3 \text{ then } n =$$

Options:

A. 3

B. 10

C. 8

D. 20

Answer: B

Solution:

Concept:

$$\bullet \quad n_{Pr} = \frac{n!}{(n-r)!} = (n-0)(n-1)(n-2)(n-3)\dots(n-r+1)$$

$$\text{For example } n_{P6} = (n-0)(n-1)(n-2)(n-3)(n-4)(n-5)$$

Calculation:

$$n_{P5} = (n-0)(n-1)(n-2)(n-3)(n-4)$$

$$n_{P3} = (n-0)(n-1)(n-2)$$

$$\text{Now given that, } n_{P5} = 42n_{P3}$$

$$\therefore (n-0)(n-1)(n-2)(n-3)(n-4) = (42)(n-0)(n-1)(n-2)$$

$$\Rightarrow (n-3)(n-4) = 42$$

$$\Rightarrow n^2 - 3n - 4n + 12 = 42$$

$$\Rightarrow n^2 - 7n - 30 = 0$$

$$\Rightarrow n^2 - 10n + 3n - 30 = 0$$

$$\Rightarrow n(n-10) + 3(n-10) = 0$$

$$\Rightarrow (n+3)(n-10) = 0$$

$$\Rightarrow n = -3 \text{ OR } n = 10$$

Well, n can't be negative,

so $n = 10$

Hence, option (2) is correct.

Question 29

Find the domain of $\sin^{-1}3x$

Options:

A. $[-1, 1]$

B. $\left[\frac{-1}{2}, \frac{1}{2}\right]$

C. $\left[\frac{-1}{3}, \frac{1}{3}\right]$

D. $\left[0, \frac{1}{3}\right]$

Answer: C

Solution:

Concept:

Principal Values of Inverse Trigonometric Functions:

Function	Domain	Range of Principal Value
$\sin^{-1}x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\csc^{-1}x$	$\mathbb{R} - (-1, 1)$	$[-\pi/2, \pi/2] - \{0\}$
$\sec^{-1}x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\pi/2\}$
$\tan^{-1}x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$
$\cot^{-1}x$	$\mathbb{R}$	$(0, \pi)$

Calculation:

Let $f(x) = \sin^{-1}3x$

As we know domain of $\sin^{-1}x$ is $x \in [-1, 1]$

So, $-1 \leq 3x \leq 1$

$$\Rightarrow \frac{-1}{3} \leq x \leq \frac{1}{3}$$

Hence domain of $\sin^{-1}3x$ is $\left[\frac{-1}{3}, \frac{1}{3}\right]$

Question 30

The area of the region bounded by the curves, $y^2 = 8x$ and $y = x$ is

Options:

A. $\frac{64}{3}$

B. $\frac{32}{3}$

C. $\frac{16}{3}$

D. $\frac{8}{3}$

Answer: B

Solution:

Given curves,

$$y^2 = 8x \rightarrow \text{(i)}$$

$$\text{and } y = x \rightarrow \text{(ii)}$$

On solving Eqs. (i) and (ii), we get

$$x^2 - 8x = 0$$

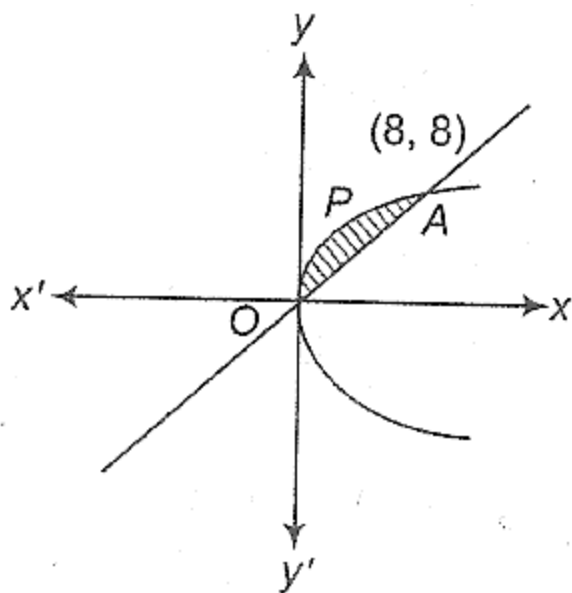
$$x(x - 8) = 0 \text{ and } y = 0, 8$$

$$\therefore \text{Required area (OPA)} = \int_0^8 (\sqrt{8x} - x) dx$$

$$= \left[2\sqrt{2} \cdot \frac{2}{3} \cdot x^{3/2} - \frac{x^2}{2} \right]_0^8$$

$$= \frac{4\sqrt{2}}{3} \cdot (8)^{3/2} - \frac{8^2}{2}$$

$$= \frac{32}{3}$$



Question 31

Solve: $\int_0^{\frac{\pi}{4}} x \cdot \sec^2 x dx$

Options:

A. $\frac{\pi}{4} + \log \sqrt{2}$

B. $\frac{\pi}{4} - \log \sqrt{2}$

C. $1 + \log \sqrt{2}$

D. $1 - \frac{1}{2} \log 2$

Answer: B

Solution:

Consider, $I = \int_0^{\frac{\pi}{4}} x \sec^2 x dx,$

Solving the indefinite integral,

,

$$I_1 = \int x \sec^2 x dx$$

,

Applying integral by parts,

,

$$\Rightarrow I_1 = x \tan x - \int \tan x dx$$

,

$$\Rightarrow I_1 = x \tan x + \ln \cos x + C$$

,

The value of the indefinite integral,

,

$$\Rightarrow I = [x \tan x + \ln \cos x]_0^{\frac{\pi}{4}}$$

,

$$\Rightarrow I = \left[\frac{\pi}{4} \tan \frac{\pi}{4} + \ln \cos \frac{\pi}{4} \right] - [0 - \ln(\cos 0)]$$

,

$$\Rightarrow I = \frac{\pi}{4} + \ln \frac{1}{\sqrt{2}}$$

Question 32

If $f(x) = \begin{cases} \frac{2x - \sin^{-1} x}{2x + \tan^{-1} x}; & x \neq 0 \\ K; & x = 0 \end{cases}$ is a continuous function at $x = 0$, then the value of K is:

Options:

A. 2

B. $\frac{1}{2}$

C. 1

D. None of these

Answer: D

Solution:

Concept:

Definition:

- A function $f(x)$ is said to be **continuous** at a point $x = a$ in its domain, if $\lim_{x \rightarrow a} f(x)$ exists or if its graph is a single unbroken curve at that point.
- $f(x)$ is continuous at $x = a \Leftrightarrow \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a} f(x) = f(a)$.

Calculation:

For $x \neq 0$, the given function can be re-written as:

$$f(x) = \begin{cases} \frac{2x - \sin^{-1} x}{2x + \tan^{-1} x}; & x \neq 0 \\ K; & x = 0 \end{cases}$$

Since the equation of the function is same for $x < 0$ and $x > 0$, we have:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0} \frac{2x - \sin^{-1} x}{2x + \tan^{-1} x}$$

$$= \lim_{x \rightarrow 0} \frac{2 - \frac{\sin^{-1} x}{x}}{2 + \frac{\tan^{-1} x}{x}} = \frac{2 - 1}{2 + 1} = \frac{1}{3}$$

For the function to be continuous at $x = 0$, we must have:

$$\lim_{x \rightarrow 0} f(x) = f(0)$$

$$\Rightarrow K = \frac{1}{3}.$$

Question 33

$$\lim_{x \rightarrow 0} \frac{\sin(2 + x) - \sin(2 - x)}{x} = ?$$

Options:

A. $\frac{1}{2} \cos 2$

B. 1

C. $2 \cos 2$

D. More than one of the above

Answer: C

Solution:

Concept:

$$\sin(a) - \sin(b) = 2 \cos\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right) \dots (1)$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \dots (2)$$

Explanation:

We are given $\lim_{x \rightarrow 0} \frac{\sin(2+x) - \sin(2-x)}{x}$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2\cos\left(\frac{(2+x)+(2-x)}{2}\right)\sin\left(\frac{(2+x)-(2-x)}{2}\right)}{x} \text{ (Using equation 1)}$$

$$\Rightarrow \lim_{x \rightarrow 0} 2\cos(2) \frac{\sin(x)}{x}$$

$$\Rightarrow 2\cos(2) \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$\Rightarrow 2 \cos(2) \times 1 \text{ (Using equation 2)}$$

$$\Rightarrow 2 \cos 2$$

Question 34

The value of $\cot \left[\cos^{-1} \left(\frac{7}{25} \right) \right]$ is

Options:

A. $\frac{25}{24}$

B. $\frac{25}{7}$

C. $\frac{24}{25}$

D. $\frac{7}{24}$

Answer: D

Solution:

Explanation:

If $\sin \theta = x \Rightarrow \theta = \sin^{-1} x$, for $\theta \in [-\pi/2, \pi/2]$

$\cot(\cot^{-1} x) = x$ for $x \in \mathbb{R}$

We have, $\cot \left[\cos^{-1} \left(\frac{7}{25} \right) \right]$

$$\text{Let } \cos^{-1}\left(\frac{7}{25}\right) = \theta$$

$$\Rightarrow \cos\theta = 7/25$$

$$\Rightarrow \sin\theta = 24/25$$

$$\Rightarrow \cot\theta = 7/24$$

$$\therefore \cot\left[\cos^{-1}\left(\frac{7}{25}\right)\right]$$

$$= \cot\theta$$

$$= 7/24$$

Question 35

What will be the value of $y''(x)$ for the given parametric form

$$x = at, \text{ and } y = a/t$$

Options:

A. $-\frac{2}{at^3}$

B. $\frac{2}{at^3}$

C. $\frac{2}{a+3}$

D. a

Answer: B

Solution:

CONCEPT:

Chain Rule of Derivatives:

- $\frac{d}{dx}f(g(x)) = \frac{d}{dg(x)}f(g(x)) \times \frac{d}{dx}g(x).$
- $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$

CALCULATION:

Given $x = at, y = a/t$

$$\Rightarrow \frac{dx}{dt} = a, \frac{dy}{dt} = \frac{a}{-t^2} \because \left(\frac{d}{dx} x^2 = nx^{n-1} \right)$$

$$\Rightarrow \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{-a}{t^2}}{\frac{a}{1}} = \frac{-a}{-at^2} = \frac{-1}{t^2}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{t^2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = y''(x) = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

$$\Rightarrow y''(x) = \frac{d}{dx} \left(-\frac{1}{t^2} \right) = \frac{d}{dt} \left(-t^{-2} \right) \frac{dt}{dx}$$

$$\Rightarrow y''(x) = \frac{2}{t^3} \times \frac{dt}{dx} = \frac{2}{t^3} \times \frac{1}{a}$$

$$\Rightarrow y''(x) = \frac{2}{at^3}$$

So, the correct answer is **option 2**.

Question 36

The value of k for which the points (2, 3), (4, k) and (5, 9) are collinear is:

Options:

A. 0

B. -7

C. 7

D. 6

Answer: C

Solution:

Formula used:

Area of a triangle:

Let coordinates be A(x₁, y₁), B(x₂, y₂), and C(x₃, y₃),

then the area of a triangle is given as,

$$\Delta = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

Note: If the point A, B, C are collinear then

The area of ΔABC should be equal to zero.

Calculation:

Given that,

(2, 3), (4, k), and (5, 9) are collinear

$$\Rightarrow \text{Area of } \Delta ABC = 0$$

$$\because \Delta = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

$$\Rightarrow \frac{1}{2} [2\{k - (9)\} + 4(9 - 3) + 5(3 - k)] = 0$$

$$\Rightarrow 2k - 18 + 24 + 15 - 5k = 0$$

$$\Rightarrow -3k + 21 = 0$$

$$\Rightarrow k = 7$$

Question 37

Find the distance between two points (2, 6, 5) and (2, 3, 9).

Options:

A. 7 units

B. 5 units

C. 0 units

D. 4 units

Answer: B

Solution:

Explanation:

The distance between two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ is

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Let A be (2, 6, 5) and B be (2, 3, 9)

$$\text{Distance AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Here, $x_1 = 2$, $y_1 = 6$, $z_1 = 5$

$x_2 = 2$, $y_2 = 3$, $z_2 = 9$

$$AB = \sqrt{(2 - 2)^2 + (3 - 6)^2 + (9 - 5)^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

= 5 units.

Question 38

If E_1 and E_2 are two events such that $P(E_1) = 0.3$, $P(E_1 \cup E_2) = 0.4$ and $P(E_2) = x$ then find the value of x such that E_1 and E_2 are two independent events ?

Options:

A. $2/7$

B. $1/5$

C. $1/7$

D. $3/5$

Answer: C

Solution:

CONCEPT:

- If A and B are two events then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- If A and B are two independent events then $P(A \cap B) = P(A) \times P(B)$

CALCULATION:

Given:

E_1 and E_2 are two independent events such that

$$P(E_1) = 0.3, P(E_1 \cup E_2) = 0.4 \text{ and } P(E_2) = x$$

Here, we have to find the value of x.

$\because E_1$ and E_2 are two independent events

As we know that,

If A and B are two independent events

$$\text{Then } P(A \cap B) = P(A) \times P(B)$$

$$\Rightarrow P(E_1 \cap E_2) = P(E_1) \times P(E_2)$$

$$\Rightarrow P(E_1 \cap E_2) = 0.3x$$

As we know that,

If A and B are two events

$$\text{Then } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$\Rightarrow 0.4 = 0.3 + x - 0.3x$$

$$\Rightarrow 0.7x = 0.1$$

$$\Rightarrow x = 1/7$$

Hence, correct option is 3.

Question 39

The inverse of the matrix $\begin{bmatrix} 2 + 3i & -i \\ i & 2 - 3i \end{bmatrix}$ is

Options:

A. $\frac{1}{12} \begin{bmatrix} 2 - 3i & i \\ -i & -2 + 3i \end{bmatrix}$

B. $\frac{1}{12} \begin{bmatrix} 2 - 3i & i \\ -i & 2 + 3i \end{bmatrix}$

C. $\frac{1}{12} \begin{bmatrix} 2 + 3i & i \\ -i & 2 - 3i \end{bmatrix}$

D. $\frac{1}{12} \begin{bmatrix} 2-3i & -i \\ i & 2+3i \end{bmatrix}$

Answer: B

Solution:

Concept:

For a 2 x 2 matrix there is a short-cut formula for inverse as given below

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{(ad-bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Calculation:

A we know that inverse of matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{(ad-bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 2+3i & -i \\ i & 2-3i \end{bmatrix}^{-1} = \frac{1}{(2+3i)(2-3i)-1} \begin{bmatrix} 2-3i & i \\ -i & 2+3i \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2+3i & -i \\ i & 2-3i \end{bmatrix}^{-1} = \frac{1}{4+9-1} \begin{bmatrix} 2-3i & i \\ -i & 2+3i \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2+3i & -i \\ i & 2-3i \end{bmatrix}^{-1} = \frac{1}{12} \begin{bmatrix} 2-3i & i \\ -i & 2+3i \end{bmatrix}$$

Hence, **option 2 is correct.**

Question 40

If chord of contact of the tangents drawn from the point (α, β) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, touches the circle $x^2 + y^2 = c^2$, then the locus of the point (α, β) is

Options:

A. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{c^2}$

B. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{c^4}$

C. $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{c^2}$

D. None of these

Answer: C

Solution:

Concept:

Equation of tangent of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

Calculation:

Given, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\Rightarrow$ Equation of tangent through (α, β) is $\frac{x\alpha}{a^2} + \frac{y\beta}{b^2} = 1 \dots (i)$

Now, (i) touches the circle $x^2 + y^2 = c^2$

$\Rightarrow$ Perpendicular distance of (i) from $(0, 0) = c$

$$\Rightarrow \frac{\frac{\alpha}{a^2}(0) + \frac{\beta}{b^2}(0) - 1}{\sqrt{\frac{\alpha^2}{a^4} + \frac{\beta^2}{b^4}}} = c$$

$$\Rightarrow \sqrt{\frac{\alpha^2}{a^4} + \frac{\beta^2}{b^4}} = \frac{-1}{c}$$

$$\Rightarrow \frac{\alpha^2}{a^4} + \frac{\beta^2}{b^4} = \frac{1}{c^2}$$

$\Rightarrow$ Locus of the points $(\alpha, \beta) : \frac{\alpha^2}{a^4} + \frac{\beta^2}{b^4} = \frac{1}{c^2}$

$\therefore$ The required answer is $\frac{\alpha^2}{a^4} + \frac{\beta^2}{b^4} = \frac{1}{c^2}$

The correct answer is Option 3.

Question 41

The distance of the point $(3, 4)$ from the line $2x + 5y + 32 = 0$, measured parallel to the line $x - y - 3 = 0$, is equal to

Options:

A. $\sqrt{2}$

B. $\sqrt{3}$

C. $\sqrt{5}$

D. $\sqrt{7}$

Answer: A

Solution:

Calculation:

Let the coordinates of P = (x, y)

Writing P in terms of parametric co-ordinates, we get:

$$x = 3 + r \cos \theta \text{ and } y = 4 + r \sin \theta \text{ and } \tan \theta = 1$$

$$\Rightarrow x = 3 + \frac{r}{\sqrt{2}} \text{ and } y = 4 + \frac{r}{\sqrt{2}}$$

It lies on $2x + 5y - 33 = 0$.

$$\Rightarrow 2\left(3 + \frac{r}{\sqrt{2}}\right) + 5\left(4 + \frac{r}{\sqrt{2}}\right) - 33 = 0$$

$$\Rightarrow 6 + \frac{7r}{\sqrt{2}} + 20 = 33$$

$$\Rightarrow \frac{7r}{\sqrt{2}} = 7$$

$$\Rightarrow r = \sqrt{2}$$

$\therefore$ The required distance is $\sqrt{2}$.

The correct answer is Option 1.

Question 42

The value (round off to one decimal place) of $\int_{-1}^1 x e^{|x|} dx$ _____

Options:

A. 1

B. 2

C. 0

D. -5

Answer: C

Solution:

Concept:

If $f(-x) = f(x)$ then the function is said to be **even function**

If $f(-x) = -f(x)$ then the function is said to be **odd function**.

For an odd function:

$$\int_{-a}^a x f(x) dx = 0$$

For an even function

$$\int_{-a}^a x f(x) dx = 2 \times \int_0^a x f(x) dx$$

Calculation:

Given:

Function $f(x) = x e^{|x|}$

Integral is -1 to 1.

Here,

$$f(-x) = -x e^{|-x|} = -x e^{|x|} = -f(x)$$

$\therefore$ The given function is an odd function.

Now, as the function is odd,

$$\int_{-1}^1 x e^{|x|} dx = 0$$

Question 43

The degree of the differential equation $\left(\frac{dy}{dx}\right)^4 + 3\frac{d^2y}{dx^2} = 0$ _____

Options:

A. Not defined

B. 2

C. 4

D. 1

Answer: D

Solution:

Concept:

Order: The order of a differential equation is the highest power of the derivative present in the equation.

Degree: The degree of a differential equation, when it is defined, is the exponent of the highest order derivative in the equation after it has been simplified as much as possible.

Solution:

The differential equation $\left(\frac{dy}{dx}\right)^4 + 3\frac{d^2y}{dx^2} = 0$

The highest-order derivative is $\frac{d^2y}{dx^2}$ whose power is 1.

Therefore, the degree of the differential equation $\left(\frac{dy}{dx}\right)^4 + 3\frac{d^2y}{dx^2} = 0$ is 1

Hence, option 4 is correct.

Question 44

What will be the sum of $3 + 7 + 11 + 15 + 19 + \dots$ upto 80 terms ?

Options:

A. 12880

B. 12400

C. 25760

D. 24800

Answer: A

Solution:

Given:

An AP is given

$3 + 7 + 11 + 15 + 19 + \dots$ upto 80 terms

Formula used:

Sum of n^{th} term of an AP

$$S_n = (n/2)\{2a + (n - 1)d\}$$

where,

'n' is Number of terms, 'a' is First term, 'd' is Common difference

Calculations:

According to the question, we have

$$S_n = (n/2)\{2a + (n - 1)d\} \text{ ----(1)}$$

where, $a = 3$, $n = 80$, $d = 7 - 3 = 4$

Put these values in (1), we get

$$\Rightarrow S_{80} = (80/2)\{2 \times 3 + (80 - 1) \times 4\}$$

$$\Rightarrow S_{80} = 40(6 + 79 \times 4)$$

$$\Rightarrow S_{80} = 40 \times 322$$

$$\Rightarrow S_{80} = 12,880$$

$\therefore$ The sum of 80^{th} terms of an AP is 12,880.

 Alternate Method

$$n^{\text{th}} \text{ term} = a + (n - 1)d$$

Here $n = 80$, $a = 3$ and $d = 4$

$$\Rightarrow 80^{\text{th}} \text{ term} = 3 + (80 - 1)4$$

$$\Rightarrow 80^{\text{th}} \text{ term} = 3 + 316$$

$$\Rightarrow 80^{\text{th}} \text{ term} = 319$$

Now, the sum of n^{th} terms of an AP

$$\Rightarrow S_n = (n/2) \times (\text{1st term} + \text{Last term})$$

$$\Rightarrow S_{80} = (80/2) \times (3 + 319)$$

$$\Rightarrow S_{80} = 40 \times 322$$

$$\Rightarrow S_{80} = 12,880$$

$\therefore$ The sum of 80th terms of an AP is 12,880.

Question 45

Solve: $5x - 3 < 3x + 1$

Options:

A. $x \leq 2$

B. $x = 2$

C. $x \geq 2$

D. $x < 2$

Answer: D

Solution:

Calculation-

$$5x - 3 < 3x + 1$$

$$\Rightarrow 2x < 4$$

$$\Rightarrow x < 2.$$

Solution of $5x - 3 < 3x + 1$ is $x < 2$.

Question 46

Find the interval in which the function $f(x) = 2x^3 - 24x + 107$ is strictly decreasing.

Options:

A. $(-\infty, -2)$

B. $(2, \infty)$

C. $[-2, 2)$

D. $(-2, 2)$

Answer: D

Solution:

Concept:

Let $y = f(x)$ be a function defined on an interval I .

Let x_1, x_2 be any two points in I , where x_1, x_2 are not the end points of the interval. Then

- $f(x)$ is a strictly increasing function if $f(x_1) < f(x_2)$ whenever $x_1 \leq x_2 \forall x_1, x_2 \in I$.

Here, $\frac{dy}{dx} > 0$ or $f'(x) > 0$

- $f(x)$ is a non-decreasing function if $f(x_1) \leq f(x_2)$ whenever $x_1 \leq x_2 \forall x_1, x_2 \in I$.

Here, $\frac{dy}{dx} \geq 0$ or $f'(x) \geq 0$

- $f(x)$ is strictly decreasing function if $f(x_1) > f(x_2)$ whenever $x_1 \leq x_2 \forall x_1, x_2 \in I$.

Here, $\frac{dy}{dx} < 0$ or $f'(x) < 0$

- $f(x)$ is a non-increasing function if $f(x_1) \geq f(x_2)$ whenever $x_1 \leq x_2 \forall x_1, x_2 \in I$.

Here, $\frac{dy}{dx} \leq 0$ or $f'(x) \leq 0$

Solution:

Given: $f(x) = 2x^3 - 24x + 27$

First find $f'(x)$.

$$\Rightarrow f'(x) = 6x^2 - 24$$

Now if $f'(x) = 0$

$$6x^2 - 24 = 0$$

$$\Rightarrow 6(x^2 - 4) = 0$$

$$\Rightarrow 6(x - 2)(x + 2) = 0$$

$$\Rightarrow (x - 2)(x + 2) = 0$$

$$\Rightarrow x = 2, -2$$

As we can see that, $f'(x) < 0$ for all $x \in (-2, 2)$

Hence, the given function is strictly decreasing $(-2, 2)$

$\therefore$ The correct option is (4)

Question 47

The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(\alpha)$ with x-axis.
The value of α is equal to

Options:

A. $\frac{\sqrt{3}}{2}$

B. $\frac{\sqrt{2}}{3}$

C. $\frac{2}{7}$

D. $\frac{3}{7}$

Answer: C

Solution:

Concept:

If there exists a line $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$, having direction ratios (a, b, c) and a plane $Ax + By + Cz + D = 0$ having direction ratios (A, B, C) then the acute angle θ between the line and the plane is given by:

$$\sin \theta = \left| \frac{aA + bB + cC}{\sqrt{a^2 + b^2 + c^2} \sqrt{A^2 + B^2 + C^2}} \right|$$

Calculation:

Given: The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(\alpha)$ with x-axis

$\therefore$ Direction ratios of the plane = (A, B, C) = (2, -3, 6)

and, direction ratios of x-axis = (a, b, c) = (1, 0, 0)

$$\therefore \sin \theta = \left| \frac{aA + bB + cC}{\sqrt{a^2 + b^2 + c^2} \sqrt{A^2 + B^2 + C^2}} \right|$$

$$= \left| \frac{1 \times 2 + 0 \times (-3) + 0 \times 6}{\sqrt{1^2 + 0^2 + 0^2} \sqrt{2^2 + (-3)^2 + 6^2}} \right|$$

$$= \left| \frac{2 + 0 + 0}{\sqrt{1 + 0 + 0} \sqrt{4 + 9 + 36}} \right|$$

$$= \frac{2}{\sqrt{49}}$$

$$= \frac{2}{7}$$

$$\Rightarrow \sin \theta = \frac{2}{7}$$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{2}{7}\right)$$

$$\therefore \alpha = \frac{2}{7}$$

The value of α is equal to $2/7$.

The correct answer is option 3.

Question 48

If the plane $2x - y + z = 0$ is parallel to the line $\frac{2x-1}{2} = \frac{2-y}{2} = \frac{z+1}{a}$, then value of a is

Options:

A. 4

B. -4

C. 2

D. -2

Answer: B

Solution:

Concept:

If a plane is parallel to a line then the normal vector to the plane will be perpendicular to the direction of the line.

The direction of the line,

$$\vec{b} = (2, 2, a)$$

The vector perpendicular to the given plane $2x - y + z = 0$,

$$\vec{n} = (2, -1, 1)$$

Calculation:

Given:

Since, the plane $2x - y + z = 0$ is parallel to the line $\frac{x-1/2}{1} = \frac{y-2}{-2} = \frac{z+1}{a}$

$$\Rightarrow 2(1) + (-1)(-2) + 1(a) = 0$$

$$\Rightarrow a + 4 = 0 \Rightarrow a = -4.$$

Question 49

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{c} = \hat{i} - 2\hat{j} + \hat{k}$, and a unit vector parallel to the vector $2\vec{a} - \vec{b} + 3\vec{c}$?

Options:

A. $\frac{9}{\sqrt{22}}\hat{i} - \frac{3}{\sqrt{22}}\hat{j} + \frac{1}{\sqrt{22}}\hat{k}$

B. $\frac{2}{\sqrt{22}}\hat{i} - \frac{3}{\sqrt{22}}\hat{j} + \frac{2}{\sqrt{22}}\hat{k}$

C. $\frac{3}{\sqrt{22}}\hat{i} - \frac{3}{\sqrt{22}}\hat{j} + \frac{3}{\sqrt{22}}\hat{k}$

D. $\frac{3}{\sqrt{22}}\hat{i} - \frac{3}{\sqrt{22}}\hat{j} + \frac{2}{\sqrt{22}}\hat{k}$

Answer: D

Solution:

Given:

$$\vec{a} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$$

$$\vec{c} = \hat{i} - 2\hat{j} + \hat{k}$$

Calculation:

The unit vector in the direction of the vector, $\vec{r} = 2\vec{a} - \vec{b} + 3\vec{c}$

$$\Rightarrow \vec{r} = 2(\hat{i} + \hat{j} + \hat{k}) - (2\hat{i} - \hat{j} + 3\hat{k}) + 3(\hat{i} - 2\hat{j} + \hat{k})$$

$$\Rightarrow \vec{r} = 2\hat{i} + 2\hat{j} + 2\hat{k} - 2\hat{i} + \hat{j} - 3\hat{k} + 3\hat{i} - 6\hat{j} + 3\hat{k}$$

$$\Rightarrow \vec{r} = (2 - 2 + 3)\hat{i} + (2 + 1 - 6)\hat{j} + (2 - 3 + 3)\hat{k}$$

$$\Rightarrow \vec{r} = 3\hat{i} - 3\hat{j} + 2\hat{k}$$

$$\text{Magnitude of } \vec{r} = \sqrt{3^2 + (-3)^2 + 2^2}$$

$$\Rightarrow |\vec{r}| = \sqrt{9 + 9 + 4} = \sqrt{22}$$

$$\text{Unit vector in the direction of } \vec{r} = \frac{1}{|\vec{r}|} \times \vec{r}$$

$$\Rightarrow \vec{r} = \frac{1}{\sqrt{22}} \times [3\hat{i} - 3\hat{j} + 2\hat{k}]$$

$$\Rightarrow \vec{r} = \frac{3}{\sqrt{22}}\hat{i} - \frac{3}{\sqrt{22}}\hat{j} + \frac{2}{\sqrt{22}}\hat{k}$$

$$\therefore \text{The required vector is } \frac{3}{\sqrt{22}}\hat{i} - \frac{3}{\sqrt{22}}\hat{j} + \frac{2}{\sqrt{22}}\hat{k}$$

Question 50

What is the square root of $z = 7 - 24i$?

Options:

A. $\pm (4 + 3i)$

B. $\pm (4i + 3)$

C. $\pm (4i - 3)$

D. $\pm (4 - 3i)$

Answer: D

Solution:

Formula used

$$\text{Roots of quadratic equation } = x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Calculation

Given $z = 7 - 24i$

Let $(a + bi)^2 = 7 - 24i$

$$\Rightarrow a^2 + 2abi + (bi)^2 = 7 - 24i$$

$$\Rightarrow a^2 - b^2 + 2abi = 7 - 24i$$

Comparing the real and imaginary parts:

$$\Rightarrow a^2 - b^2 = 7 \dots\dots(1)$$

$$\Rightarrow 2ab = -24 \dots\dots(2)$$

$$\Rightarrow b = \frac{-24}{2a} = \frac{-12}{a}$$

Substitute b into (1)

$$\Rightarrow a^2 - \left(\frac{-12}{a}\right)^2 = 7 \Rightarrow a^2 - \frac{144}{a^2} = 7$$

$$\Rightarrow a^4 - 144 = 7a^2$$

Let $x = a^2$ Then the equation becomes:

$$\Rightarrow x^2 - 7x - 144 = 0$$

$$\Rightarrow x = \frac{7 \pm \sqrt{49 + 576}}{2} \Rightarrow x = \frac{7 \pm 25}{2}$$

$x = 16$, $x = -9$ (This is not valid since $x = a^2$ cannot be negative)

$$\text{So, } a^2 = 16 \Rightarrow a = \pm 4$$

$$\text{When } a = 4, b = \frac{-12}{4} = -3$$

$$\text{Or if } a = -4, b = \frac{-12}{-4} = 3$$

Thus, the two possible roots are $(4 - 3i)$ and $(-4 + 3i)$.

$\therefore$ The correct answer is: **Option 4) $\pm (4 - 3i)$**

English

Question 51

Choose the word which is similar in meaning to the italicized word in this sentence.

He could rise to this stature because of his *invincible* courage.

Options:

- A. Susceptible
- B. Unmanageable
- C. Unbeatable
- D. Immeasurable

Answer: C

Solution:

The correct answer is '**Unbeatable**'.



Key Points

- **invincible** means too powerful to be defeated or overcome.
 - **Synonyms** are invulnerable, indestructible, unconquerable, unbeatable, etc.
 - Example: "an **invincible** warrior"
- **Unbeatable** means are not able to be defeated or bettered in a contest or commercial market.
- After reading the word which is similar in meaning to the italicized word in this sentence is '**Unbeatable**'.

Therefore, the correct answer is **Option 3**.



Additional Information

- **Susceptible**: likely or liable to be influenced or harmed by a particular thing.
- **Unmanageable**: difficult or impossible to manage, manipulate, or control.
- **Immeasurable**: too large, extensive, or extreme to measure.

Question 52

Directions: Each item in this section consists of a sentence with an underlined word followed by four words/groups of words. Select the option that is nearest in meaning to the underlined word and mark your response on the Answer Sheet accordingly.

Their discussion was "fruitful", leading to several actionable ideas by the end of the session.

Options:

A. unsuccessful

B. productive

C. useless

D. lengthy

Answer: B

Solution:

The correct answer is: **Productive**.

Key Points

- The word **"fruitful"** means producing good or helpful results; productive. (फलदायक)
 - **Example:** The research team had a fruitful meeting, resulting in a new direction for their project.
- **"Productive"** also refers to producing or able to produce large amounts of goods, crops, or other commodities; it denotes effectiveness in bringing about results. (उत्पादक)
 - **Example:** She had a very productive day at work, finishing tasks that had been pending for weeks.
- Hence, we can infer that the nearest meaning of **'fruitful'** is **'productive'**.

Therefore, the correct answer is: **Productive**.

Additional Information

Here are the other options explained along with their Hindi meanings and example sentences:

- **Unsuccessful** (असफल): Not achieving the desired aims or results.
 - **Example:** Despite their best efforts, the team was unsuccessful in winning the championship.
- **Useless** (बेकार): Not fulfilling or not expected to achieve the intended purpose or desired outcome.
 - **Example:** He felt that the meeting was completely useless and a waste of time.
- **Lengthy** (लंबा): Something that is longer than usual or necessary.
 - **Example:** The legal process was lengthy and drained their resources.

Question 53

Choose the correct spelling:

Options:

- A. ambiguous
- B. ambiguos
- C. ambiguous
- D. ambeeguuous

Answer: A

Solution:

The correct spelling is '**ambiguous**'.



Key Points

- '**Ambiguous**' means having more than one possible meaning.
 - **Example**-*Her songs are intentionally **ambiguous**.*
 - Option 2, 3 and 4 are incorrect.
 - Hence, the correct answer is **option 1**.
-

Question 54

Which of the following words is correctly spelt?

Options:

- A. Vaccum
- B. Vacuum
- C. Vaacum
- D. Vacuum

Answer: D

Solution:

The correct answer is '**Vacuum**'.

Key Points

- The word '**vacuum**', spelled with **single 'c'** and **double 'u'**, is a commonly misspelled word.
- **Vacuum** is a space that is completely empty of all substances, including air or other gases.
- It also means a situation from which something is missing or lacking or the act of cleaning something with a vacuum cleaner.
- Hence, **option 4 i.e. 'vacuum'** is the correct answer.

Additional Information

- The word '**vacuum**' is also used as a **verb** which means to clean something using a vacuum cleaner.
 - **For eg.-** The maid **vacuumed** the floor twice a week.
-

Question 55

Choose the most appropriate answer and fill in the blanks:

The plural of 'sheep' is _____.

Options:

- A. Sheeps
- B. Sheep
- C. Sheepes
- D. None of the above

Answer: B

Solution:

The plural of 'sheep' is 'sheep'.

There are a number of nouns whose plural is the same as the singular.

Therefore, the correct answer is option 2.

Question 56

Direction: The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer.

No sooner he was / brought to the hospital / than he began panting heavily.

Options:

- A. brought to the hospital
- B. No sooner he was
- C. No error
- D. than he began panting heavily

Answer: B

Solution:

Here, the error lies in the part "**No sooner he was.**"

Key Points

- The error is the use of "**he was**" after "**no sooner.**"
- "**No sooner**" is a conjunction and requires an inversion of the subject and auxiliary verb.
- The correct structure is "**No sooner + auxiliary verb + subject + verb**" or "**No sooner + subject + auxiliary verb + verb.**"
 - **Example:** *No sooner had she finished her speech than the lights went out.*
- The correct structure is "*No sooner was he brought to the hospital*" or "*He was no sooner brought to the hospital than he began panting heavily.*"
- So, the correct answer is Option 2.

Question 57

The following sentence has been split into four segments. Identify the segment that contains a grammatical error.

She will have participate / in the dance competition / by the time her father / returns from the tour.

Options:

- A. in the dance competition
- B. She will have participate
- C. by the time her father
- D. returns from the tour

Answer: B

Solution:

The erroneous part is 'She will have participate'.



Key Points

- The segment that contains a grammatical error is: **She will have participate.**
- The correct form of the verb should be in the **past participle form, which is "participated."** The corrected segment should read: **"She will have participated."**
- Explanation: The verb **"participate"** is incorrectly used in the **present tense form "participate"** instead of the **past participle form "participated"** following the **auxiliary verb "have."**
- In this context, the **past participle form** is needed to convey the completion of the action.

Correct Sentence: *She will have participated in the dance competition by the time her father returns from the tour.*

Question 58

Isn't it strange that though we live in the same city, I get to see my sister once in a blue moon?

What does 'once in a blue moon' mean?

Options:

- A. Everyday
- B. All the time

C. Frequently

D. Occasionally

Answer: D

Solution:

The correct answer is '**Occasionally**'.



Key Points

- '**Once in a blue moon**' is an idiomatic phrase that means **occasionally or very rarely**.
 - **Example:** Peter only comes out for a drink **once in blue moon** now that he has kids.
 - The sentence means - *Isn't it strange that though we live in the same city, I get to see my sister occasionally?*
 - Thus, from the above given explanation the correct answer is option 4.
-

Question 59

Select the option that can be used as a one-word substitute for the given group of words.

An apparatus used by submerged submarines to see above water

Options:

A. Kaleidoscope

B. Telescope

C. Periscope

D. Bioscope

Answer: C

Solution:

The correct answer is: **Periscope**.



Key Points

- The term "**apparatus**" refers to the tools or equipment needed for a specific purpose, especially for scientific or technical activities. (उपकरण)
 - **Example:** The laboratory was equipped with all the latest apparatus for conducting advanced research.
- "**Periscope**" is a device used for observation over, around, or through an object, obstacle, or condition that prevents direct line-of-sight observation from the observer's current position. In the context of submarines, it is used to see above water while being submerged. (पेरिस्कोप)
 - **Example:** The submarine's captain used the periscope to survey the surface of the ocean without emerging from the depths.
- Hence, we can infer that an apparatus used by submerged submarines to see above water is a '**Periscope**'.

Therefore, the correct answer is: **Periscope**.

Additional Information

Here are the other options explained along with their Hindi meanings and example sentences:

- **Kaleidoscope** (जलतरंग): A tube containing mirrors and pieces of colored glass or paper, whose reflections produce changing patterns that are visible through an eyehole when the tube is rotated.
 - **Example:** The child was fascinated by the changing patterns of the kaleidoscope.
- **Telescope** (दूरबीन): An optical instrument designed to make distant objects appear nearer, containing an arrangement of lenses, or of curved mirrors and lenses, by which rays of light are collected and focused and the resulting image magnified.
 - **Example:** The astronomer used a telescope to observe the stars and planets.
- **Bioscope** (बायोस्कोप): Historically, a bioscope was an early form of projector or cinema capable of showing moving pictures. The term is now obsolete.
 - **Example:** In the early 20th century, people flocked to bioscopes to catch a glimpse of the marvels of motion pictures.

Question 60

Which term refers to risking a sum of money or valued item(s) on the outcome of a future event?

Options:

- A. Whorl
- B. Writhe
- C. Wager
- D. Warden

Answer: C

Solution:

The correct answer is 'Option 3) Wager'.

Key Points

- The given phrase describes wagering.
- 'Wager' in English means to risk a sum of money or valued item(s) on the outcome of a future event and in Hindi, it's termed as "शर्त".
- For example- "He made a wager on the outcome of the game."
- Wager refers to an act of betting a certain amount on something uncertain.

Therefore, the correct answer is- Correct Answer: 3) Wager.

Additional Information

- Option 1) 'Whorl' refers to a pattern of spirals or concentric circles. (घुंघराला).
 - Option 2) 'Writhe' means to make continuous twisting, squirming movements, or contortions of the body. (मरोड़ना)
 - Option 4) 'Warden' refers to a person responsible for the supervision of a particular place or thing or for enforcing the regulations associated with it. (प्रबंधक)
-

Question 61

Direction: Use the right preposition.

He deals _____ cosmetic goods.

Options:

A. of

B. with

C. in

D. on

Answer: C

Solution:

The correct answer is **option 3) i.e. in.**

Key Points

- Let us explore the given options:
 - The preposition '**of**' is used to indicate relating to, belonging to.
 - The preposition '**with**' means having or holding something.
 - The preposition '**in**' is used while doing something.
 - The preposition '**on**' to refer a state.
- Hence, the correct answer is '**in**'.

Complete Sentence: *He deals **in** cosmetic goods.*

Additional Information

- A **preposition** is a word that indicates the relationship between a noun and the other words of a sentence. They explain relationships of sequence, space, and logic between the object of the sentence and the rest of the sentence. They help us understand order, time connections, and positions.
-

Question 62

Directions: Each of the following sentences in this section has a blank space with four options. Select whichever preposition or determiner you consider the most appropriate for the blank space and indicate your response on the Answer Sheet accordingly.

We had a discussion ____ the project.

Options:

- A. beneath
- B. with
- C. about
- D. in

Answer: C

Solution:

The correct answer is **Option 3** i.e. **about**.

Key Points

- The sentence "We had a discussion ____ the project." requires a preposition that indicates the subject of the discussion.
- The word "**about**" is the correct choice because it shows that the discussion was focused on the project.
- The options "beneath," "with," and "in" do not correctly complete the sentence to convey the intended meaning of discussing the topic of the project.
- Using "about" correctly specifies the topic under discussion, making it clear that the project is what was being discussed.

Therefore, the **correct answer** is- **about**.

Additional Information

- **Beneath** means at a lower level or layer.
- **With** is used to indicate the instrument, manner, or agent of an action.
- **In** is used to indicate inclusion within space, a place, or limits.

Question 63

Select the most appropriate option to fill in the blank.

As the school year was about to end, the students _____ their teacher a special present _____ their affection.

Options:

- A. presented; to show
- B. is presenting; showing
- C. were presented; to show
- D. present; showed

Answer: A

Solution:

The correct answer is "**presented; to show**."

Key Points

- "**Presented**": This is the past tense of the verb 'present'.
- It indicates that the action of giving a present has already occurred.
- The sentence's context ("**As the school year was about to end**") makes it clear that this is a past event.
- Therefore, the past tense verb '**presented**' is the appropriate choice.
- "**To show**": This phrase is used to explain why the students presented their teacher with a special present.

- The infinitive '**to show**' expresses the purpose or intent behind their action. In this case, the purpose was to manifest (show) their affection to their teacher.
- The correctly completed sentence is:
 - "**As the school year was about to end, the students presented their teacher a special present to show their affection.**"

Therefore, the correct answer is "**Option 1**".

Question 64

Select the correct indirect form of the given sentence.

Nakul asked her, "How do you know that Rahul has told that to him?"

Options:

- A. Nakul asked her how she knew that Rahul had told that to him.
- B. Nakul asked her that how she has known that Rahul has told that to him.
- C. Nakul asked her how she knew that Rahul has told that to him.
- D. Nakul asked her that how she knows that Rahul had told that to him.

Answer: A

Solution:

The correct answer is '**Nakul asked her how she knew that Rahul had told that to him.**'

Key Points

- The given sentence is in **Direct Speech**. As per the question we have to change it into **Indirect Speech**.
- The process of transformation as follows:
 - '**Asked**' will not change because the given sentence is an '**Interrogative sentence**'.
 - '**How**' will be used as a conjunction because we know that in an **interrogative sentence with a question word the question word itself is used as a conjunction**.
 - '**You**' will be changed into '**she**' because second-person pronouns are changed according to the **reporting verb** of the sentence in '**direct to indirect**' transformation.
 - We know that in the '**direct to indirect**' transformation '**First form**' of the verb is changed into '**Second form**' of the verb.
 - '**Has told**' will be changed into '**Had told**'.
- Therefore, as per the points mentioned above, we find that the correct answer is **Option 1**.

Correct Sentence: *Nakul asked her how she knew that Rahul had told that to him.*

Additional Information

- Transformation of Imperative Sentence:
 - 'Orders', 'Command', 'Request', 'Advice' are given in these kinds of sentences.
 - Reporting Verb is changed into **ordered, advised, requested, commanded, beg, forbade, suggested, proposed, assured, asked, reminded, warned, agreed, refused, promised**, etc.
 - Inverted Commas will be removed. And conjunction **'to/not to'** will be used.
 - Example:
 - He said, "John, please bring me my towel." (Direct Speech)
 - He requested John to bring him his towel. (Indirect Speech)
-

Question 65

Select the most appropriate indirect form of the given sentence.

“I finished my assignment two hours ago,” she said.

Options:

- A. She said that she finished my assignment two hours ago.
- B. She said that she had finished her assignment two hours before.
- C. She said that I finished my assignment two hours before.
- D. She said that she has finished her assignment two hours ago.

Answer: B

Solution:

The indirect speech of the given sentence is **She said that she had finished her assignment two hours before.**

Key Points

She said I finished my assignment two hours ago.
--

Reporting clause Reported speech

- When the reporting speech is in **Simple Past tense** then all the speech changes to **past perfect tense**.
 - Finished-----had finished**
- Said** of the reporting verb remains the same.
- Replace **comma (,) and inverted commas (" ")** with the conjunction **'that'**.
- I (first person)** of the Reported Speech changed according to the subject of the Reporting clause (**She**).
- Ago** changed into **Before**.

So, the final sentence is: She said that she had finished her assignment two hours before.

Additional Information

What is Direct & Indirect Speech?

- **Direct speech**— reporting the message of the speaker in the exact words as spoken by him.
 - **Direct speech example:** Maya said, "I am busy now".
 - **Indirect speech:** reporting the message of the speaker in our own words
 - **Indirect speech example:** Maya said that she was busy then.
-

Question 66

Select the correct passive form of the given sentence.

Humayun built a city named Dinpanah in 1533 in Delhi.

Options:

- A. A city named Dinpanah was built by Humayun in 1533 in Delhi.
- B. A city named Dinpanah had been built by Humayun in 1533 in Delhi.
- C. A city named Humayun was built by Dinpanah in 1533 in Delhi.
- D. A city named Dinpanah was being built by Humayun in 1533 in Delhi.

Answer: A

Solution:

The correct answer is **Option 1)**

Key Points

- The given sentence is in active voice so, we have to change it in passive voice.
- In the given sentence, the verb '**built**' is correctly changed to '**was built**' to form the passive voice.
- The subject '**Humayun**' is moved to the end of the sentence and preceded by '**by**'.
- The object '**a city named Dinpanah**' becomes the subject of the **passive sentence**.

Hence, the correct statement is '**A city named Dinpanah was built by Humayun in 1533 in Delhi.**'

Additional Information

- Other given options are not correct as they are not in correct format.

Question 67

Select the correct active voice of the given sentence

Why was such a letter written by Rupesh?

Options:

- A. Why did Rupesh wrote such a letter?
- B. Why has Rupesh writes such a letter?
- C. Why is Rupesh write such a letter?
- D. Why did Rupesh write such a letter?

Answer: D

Solution:

The correct answer is **Option 4**.



Key Points

- In passive voice, the sentence is: "**Why was such a letter written by Rupesh?**"
- To convert to active voice, the subject "**Rupesh**" should come first, followed by the verb and the object.
- Ensure the question format is maintained with appropriate auxiliary verb "did" for past tense questions.
- Correct verb form should be used; "**written**" changes to "write" to match the base form after "did."
- Other options use incorrect verb forms ("wrote" instead of "write") or incorrect auxiliary verb.

Therefore, the correct answer is **Option 4**.

Correct Sentence: "*Why did Rupesh write such a letter?*"



Additional Information

- **Option 1:** "Why did Rupesh wrote such a letter?" – Incorrect verb form "wrote" after "did."
- **Option 2:** "Why has Rupesh writes such a letter?" – Incorrect auxiliary verb "has" and verb form "writes."
- **Option 3:** "Why is Rupesh write such a letter?" – Incorrect auxiliary verb "is" and verb form "write."

Question 68

Read the following information carefully and answer the given questions.

As humans we really don't think that psychologically, birds are capable of the same mental tasks that other mammals are, and much more specifically, some of them are in the rarest of company with us, they are near-sapient, just like our close relatives.

Birds have been shown in previous studies to possess a range of skills such as complex social reasoning, an ability to problem solve and some have even demonstrated the capability to craft and use tools. Both birds and mammals are warm-blooded, which means they can maintain a constant body temperature and do not need to rely on an external heat source to stay warm. This similarity lends itself to several other commonalities, such as similar caloric requirements by weight and the ability to remain active in colder temperatures. Cold-blooded animals do not have to eat as much, but they also cannot survive colder temperatures.

Birds require a lot of energy in order to fly. This also necessitates a circulatory system that is both efficient and effective, so they have evolved a four-chambered heart with two atria and two ventricles, just like mammals.

The blood of birds and mammals contains both red and white blood cells, called erythrocytes and leukocytes respectively. The red blood cells in both classes of animals contain haemoglobin. Another similarity between birds and mammals is that both classes care for their young after they're hatched or born. Female mammals feed their young by lactating, while birds feed their young beak to beak.

What is the main purpose of the passage?

Options:

- A. To encourage scientists to research more on birds.
- B. To understand how birds and mammals are alike.
- C. To explain the differences between birds and humans.
- D. To motivate people to interact more with birds.

Answer: B

Solution:

The correct answer is '**To understand how birds and mammals are alike**'.



Key Points

- The passage is primarily discussing different aspects where birds and mammals share similarities such as maintaining constant body temperature, possessing red and white blood cells, care for their offspring, and more.
- Option 1, which encourages scientists to research more on birds, is not the main theme, as the passage does not actively promote or request further research on the subject of birds.
- While the passage does discuss some comparisons with humans, it isn't mainly focused on the differences between birds and humans, as suggested by Option 3.
- The author does not motivate people to interact more with birds (Option 4), rather the focus is on the shared characteristics between birds and mammals.

Therefore, the correct answer is Option 2, 'To understand how birds and mammals are alike'.

Question 69

Read the following information carefully and answer the given questions.

As humans we really don't think that psychologically, birds are capable of the same mental tasks that other mammals are, and much more specifically, some of them are in the rarest of company with us, they are near-sapient, just like our close relatives.

Birds have been shown in previous studies to possess a range of skills such as complex social reasoning, an ability to problem solve and some have even demonstrated the capability to craft and use tools. Both birds and mammals are warm-blooded, which means they can maintain a constant body temperature and do not need to rely on an external heat source to stay warm. This similarity lends itself to several other commonalities, such as similar caloric requirements by weight and the ability to remain active in colder temperatures. Cold-blooded animals do not have to eat as much, but they also cannot survive colder temperatures.

Birds require a lot of energy in order to fly. This also necessitates a circulatory system that is both efficient and effective, so they have evolved a four-chambered heart with two atria and two ventricles, just like mammals.

The blood of birds and mammals contains both red and white blood cells, called erythrocytes and leukocytes respectively. The red blood cells in both classes of animals contain haemoglobin. Another similarity between birds and mammals is that both classes care for their young after they're hatched or born. Female mammals feed their young by lactating, while birds feed their young beak to beak.

The author has discussed which of the following skills of birds in the passage?

I. Complex social reasoning

II. Problem solving skill

III. Capability to craft tools

Options:

A. Only III

B. I and II

C. II and III

D. I, II and III

Answer: D

Solution:

The correct answer is 'I, II and III-Complex social reasoning, Problem-solving skill, and Capability to craft tools'.



Key Points

- All three skills - 'complex social reasoning', 'problem-solving skills', and 'capability to craft and use tools' - are mentioned in the passage as abilities possessed by birds.
 - These skills emphasize birds' cognitive abilities, putting them on par with some mammals, including humans, in terms of certain mental tasks.
 - Therefore, the correct answer is Option 4, 'Complex social reasoning, Problem-solving skill, and Capability to craft tools'.
-

Question 70

Read the following information carefully and answer the given questions.

As humans we really don't think that psychologically, birds are capable of the same mental tasks that other mammals are, and much more specifically, some of them are in the rarest of company with us, they are near-sapient, just like our close relatives.

Birds have been shown in previous studies to possess a range of skills such as complex social reasoning, an ability to problem solve and some have even demonstrated the capability to craft and use tools. Both birds and mammals are warm-blooded, which means they can maintain a constant body temperature and do not need to rely on an external heat source to stay warm. This similarity lends itself to several other commonalities, such as similar caloric requirements by weight and the ability to remain active in colder temperatures. Cold-blooded animals do not have to eat as much, but they also cannot survive colder temperatures.

Birds require a lot of energy in order to fly. This also necessitates a circulatory system that is both efficient and effective, so they have evolved a four-chambered heart with two atria and two ventricles, just like mammals.

The blood of birds and mammals contains both red and white blood cells, called erythrocytes and leukocytes respectively. The red blood cells in both classes of animals contain haemoglobin. Another similarity between birds and mammals is that both classes care for their young after they're hatched or born. Female mammals feed their young by lactating, while birds feed their young beak to beak.

Which characteristic of birds show that they are caring?

Options:

- A. They fly together in a particular area.
- B. They don't leave the side of their kind.
- C. They feed their young beak to beak.

D. They have the capability to use tools.

Answer: C

Solution:

The correct answer is '**They feed their young beak to beak**'.

Key Points

- The passage clearly mentions that both birds and mammals care for their young once they are born or hatched. In the case of birds, this nurturing characteristic is exemplified by how they feed their young beak to beak.
- While other options might suggest certain behavioral traits of birds, they aren't explicitly mentioned in the passage nor are these traits necessarily indicative of a 'caring' nature as specified by the question.

Therefore, the correct answer is Option 3, 'They feed their young beak to beak'.
